

Are galaxies in compact groups special?

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ABSTRACT

Context. Compact groups combine high projected densities with low velocity dispersions, but many are embedded in or projected onto larger structures. Their evolutionary significance depends on whether their galaxies remain distinct from regular-group analogues, not only from field galaxies.

Aims. We test whether CG₄ galaxies remain special after removing split systems and comparing them with ordinary four-member groups, luminous regular-group cores, and projected compact cores around brightest group galaxies. We ask which observable carries any residual signal: morphology, star-formation state, colour, AGN-like activity, group-scale structure, phase-space position, or local projected tidal density.

Methods. We use the CG₄ sample inherited from the Tricottet et al. construction, leaving 62 compact groups after the Zheng & Shen split-system cut, and compare it with RG₄, Control_{4B}, and Control_{4C}. We combine Galaxy Zoo vote-fraction morphologies, SDSS stellar masses and specific star-formation rates, optical colours, emission-line diagnostics, adjusted binomial models, and one-to-one matched controls.

Results. CG₄ galaxies are more often elliptical/smooth and less often spiral/features than control galaxies. This morphology signal survives adjustment for stellar mass, redshift, rank, and available group-scale quantities, and is also present in matched comparisons. The star-formation signal is weaker: CG₄ satellites have a lower raw star-forming fraction, especially relative to RG₄, but this contrast is not comparably robust after adjustment and matching. Among galaxies that remain star-forming, we find no robust compact-group offset from the star-forming main sequence. Colours, AGN-like fractions, magnitude gaps, and the limited H α diagnostics do not support a simple instantaneous-quenching, fossil-group, or AGN-driven explanation. A projected pairwise tidal-index diagnostic suggests that local tidal density accounts for part of the morphology signal.

Conclusions. Compact groups host a more morphologically evolved population, consistent with accumulated transformation in dense group cores and/or preprocessing. The evidence favours a cumulative structural or morphological signal over a uniquely isolated compact-group quenching mode at the present epoch.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: elliptical and lenticular, cD – methods: statistical

1. Introduction

Compact groups of galaxies are useful laboratories for environmental evolution because they combine high projected galaxy densities with relatively low velocity dispersions (Hickson 1982; Hickson et al. 1992). In such configurations, repeated tidal encounters, dynamical friction, mergers, gas stripping or consumption, and disk heating can plausibly accelerate the transformation of galaxies compared with more diffuse environments (Barnes 1985, 1989; Verdes-Montenegro et al. 2001). This expectation is closely connected to the broader morphology-density and star-formation-environment relation, in which galaxy structure and star-formation activity vary systematically with local density (Dressler 1980).

The interpretation of compact groups is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities. Simulations and redshift-space anal-

yses have long shown that apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2012; Zheng & Shen 2021). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints.

Recent structural work strengthens this motivation. Using S-PLUS imaging and single-Sérsic measurements, Montaguth et al. (2023, 2025, 2026b,a) identified transition galaxies and trends in the R_e -Sérsic-index plane that point to structural transformation in compact groups, with stronger signatures in non-isolated compact groups and host structures. These studies suggest that compact groups can mark sites of cumulative structural processing. They do not, however, by themselves answer whether compact-group

galaxies remain distinct after comparison with regular-group cores selected from the same parent catalogues.

Our previous sample-construction paper (Tricottet et al. 2025) showed that many compact groups are not isolated systems, but appear as dense cores or substructures of richer environments. This result motivates the stricter comparison made here. We use the same compact-group and regular-group sample framework to ask whether CG galaxies are still special after controlling for stellar mass, redshift, galaxy rank, and available group-scale properties. We then ask which observable carries the signal: star-formation class, Galaxy Zoo morphology, optical colour, AGN-like activity, fossil-like magnitude gaps, projected phase-space position, or local projected tidal density.

The paper is organized as follows. Section 2 describes the samples, the sSFR and morphology classifications, and the statistical hierarchy. Sections 3 and 4 present baseline population differences and group-scale diagnostics. Section 5 tests which signals survive adjusted and matched comparisons and examines secondary diagnostics. Section 6 discusses the physical interpretation of the morphology signal, and Section 7 summarizes the conclusions.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study Tricottet et al. (2025). We briefly recall their construction here. Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018). Starting from the reliable HMCs with exactly four members, we retain only systems in which all galaxies are present in the Lim–Tempel catalogue and impose the volume- and luminosity-complete limits $0.005 < z_{\text{BGG}} < 0.0452$ and $M_{r,\text{BGG}} \leq -21.81$, with all 4 galaxies lying within 3 magnitudes of the brightest group galaxy. This defines the parent CG₄ sample of 78 compact groups. The regular-group comparison samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least 4 members and at least 3 satellites within 3 magnitudes of the BGG. From this parent control catalogue, we define RG₄ groups as regular groups with exactly four members, excluding systems identical to a CG₄, and we additionally use the Control_{4B} and Control_{4C} samples, formed respectively from the 4 brightest companions in each parent regular group, after removing groups that share galaxies with CG₄s. In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021). This reduces the compact-group sample from 78 to 62 groups. We also remove the systems associated with Lim group 3688 from our Control_{4B} and Control_{4C} samples. This group contains 6 galaxies with redshifts between 0.032 and 0.034, and 1 with a redshift of 0.048. This group appears to be spurious, and the outlying galaxy, although not the BGG, is selected in both our original Control_{4B} and Control_{4C} samples, significantly af-

fecting the group velocity dispersion (1979 and 1981 km s⁻¹, respectively).

The three regular-group controls serve different physical purposes. RG₄ contains true regular groups with exactly four selected members and is the most direct four-member regular-group comparison. Control_{4B} contains the four brightest galaxies in richer regular groups and tests whether the CG₄ population resembles the luminous core of an ordinary group. Control_{4C} contains the BGG plus the three closest projected companions and is the most compact-core-like ordinary-group analogue. CG₄ is therefore tested against both equal-membership groups and regular-group cores, rather than against a single control definition.

We use SDSS DR 16 to retrieve stellar masses, star formation rates (SFRs), and Galaxy Zoo morphologies (Lintott et al. 2011). We specifically extract the MPA-JHU galSpecExtra columns `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50`, whose stellar-mass and star-formation estimates follow Kauffmann et al. (2003) and Brinchmann et al. (2004), and `p_el_debiased` and `p_cs_debiased` from the zooSpec table. Projected physical distances that are recomputed from angular separations use the Planck15 cosmology (Planck Collaboration et al. 2016); published catalogue quantities are used as tabulated, and no additional IMF or *h*-rescaling is applied.

The number of usable galaxies varies slightly among the downstream tables because not every object has every required SDSS-derived quantity. Table 1 summarizes the parent counts and the main complete-case subsets used below.

2.2. sSFR classification

To build a general star formation classification that could be uniformly applied to all our samples, we extracted galaxies from the SDSS DR 16 using the query shown in Listing 1.

```

SELECT
  s.specObjID,
  s.z,
  p.petroMag_r - p.extinction_r as
    r_obs,
  p.petroMag_u - p.extinction_u as
    u_obs,
  p.petroMag_g - p.extinction_g as
    g_obs,
  p.petroMag_i - p.extinction_i as
    i_obs,
  p.petroMag_z - p.extinction_z as
    z_obs,
  p.objID,
  g.sfr_tot_p50, g.specsfr_tot_p50, g.
    lgm_tot_p50,
  l.h_alpha_eqw, l.h_beta_eqw, l.
    oiii_5007_eqw, l.nii_6584_eqw,
  l.h_alpha_flux, l.h_beta_flux, l.
    oiii_5007_flux, l.nii_6584_flux,
  z.p_el_debiased AS p_E,
  z.p_cs_debiased AS p_S
FROM SpecObj AS s
JOIN PhotoObj AS p ON s.bestObjID = p
  .objID
JOIN galSpecExtra as g ON s.specObjID
  = g.specObjID

```

Sample	Parent	sSFR	Morph.	Secure morph.	Mass	Colours	Sat.-BGG dist.	Matched	BPT
CG ₄	248	248	248	210	234	217	186	234	184
Control _{4B}	2792	2792	2792	2372	2641	2488	2094	164	2321
Control _{4C}	3004	3004	3004	2464	2839	2691	2253	70	2325
RG ₄	224	224	224	189	219	204	168	–	189

Table 1. Sample-size audit for the main analysis subsets. “Secure morph.” counts galaxies classified as either elliptical/smooth or spiral/features rather than uncertain. “Sat.-BGG dist.” counts satellites with a projected BGG distance. “Matched” is the number retained in the one-to-one matched-control analysis, where applicable. “BPT” counts galaxies with the diagnostic line ratios needed for the emission-line classification.

```

19 JOIN galSpecLine as l ON s.specObjID
20 = l.specObjID
21 JOIN zooSpec AS z ON s.specObjID = z.
22 specObjID
23 WHERE s.z BETWEEN 0.005 AND 0.0452
24 AND (p.petroMag_r - p.extinction_r <=
1703 AND s.class = 'GALAXY'
24 AND g.lgm_tot_p50 > -1000

```

Listing 1. Query used for selecting spectral & photometric data from the SDSS

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\text{NII}]\lambda 6584/\text{H}\alpha) \text{ and } \log_{10}([\text{OIII}]\lambda 5007/\text{H}\beta).$$

Among these galaxies, we flag broad AGN-like systems for exclusion from the GMM calibration sample if they satisfy either of the following conditions:

$$\log_{10} \frac{[\text{NII}]}{\text{H}\alpha} > 0,$$

or if they lie above the empirical demarcation of Kauffmann et al. (2003):

$$\log_{10} \frac{[\text{OIII}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{NII}]/\text{H}\alpha) - 0.05} + 1.3.$$

Galaxies satisfying this broad flag are excluded from the GMM calibration sample. The later emission-line diagnostic uses the usual BPT classes: star-forming below the Kauffmann et al. (2003) curve, composite between Kauffmann et al. (2003) and Kewley et al. (2001), and AGN-like above the Kewley et al. (2001) maximum-starburst line or already carrying the catalogue AGN flag. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN. They are retained for the broad sSFR status assignment when valid SDSS stellar mass and sSFR estimates are available, while the emission-line/AGN-like analysis below treats BPT-unclassified objects separately. The BPT-classifiable subset is shown in Fig. 1.

Galaxies with `specsfr_tot_p50` equal to -9999 in the SDSS are classified as quenched. In figures that require a continuous ordinate, these objects are shown at the arbitrary plotting floor $10^{-14.0} \text{ yr}^{-1}$; this value is used only for display and is not interpreted as a measured sSFR or used in the fitted comparisons.

The remaining galaxies are separated into star-forming and passive populations with a two-component Gaussian Mixture Model (GMM) in the $\log M_*$ – $\log \text{sSFR}$ plane. The model is calibrated on galaxies not flagged as AGN-like

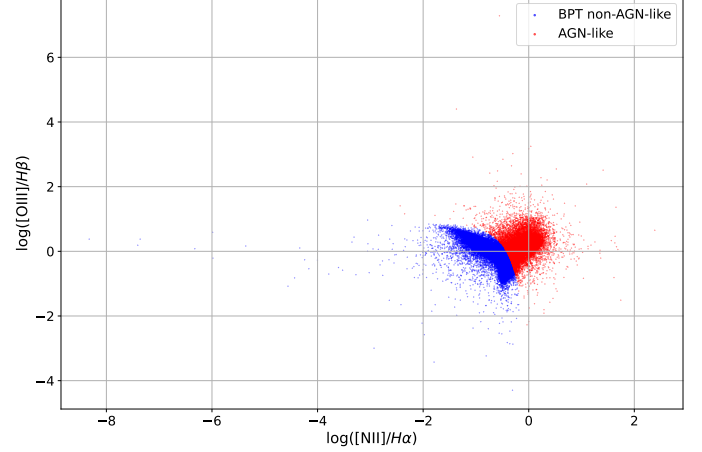


Fig. 1. BPT diagram for galaxies with measured diagnostic line ratios. The AGN-like region is excluded from the sSFR calibration sample; galaxies without the required ratios are BPT-unclassified rather than classified as non-AGN by this diagram.

when the required BPT ratios are available, while BPT-unclassified galaxies with valid SDSS stellar mass and sSFR estimates can still receive a broad sSFR class. The GMM provides a data-driven division between the high-sSFR sequence and the low-sSFR cloud, in the spirit of previous multi-Gaussian decompositions of the $\text{SFR}-M_*$ plane (McLachlan & Peel 2000; Wuyts et al. 2011; Hahn et al. 2019). We use the equal-posterior boundary between the two components as the passive/star-forming separator. The exact optimizer settings are not used in the interpretation; the key results are checked below against a simpler fixed-sSFR threshold.

2.3. Raw sSFR fractions

The sSFR status for each sample is shown in Table 2. In the raw all-galaxy comparison, Fisher’s exact test gives star-forming-fraction p-values of $p = 0.648$ versus the Control_{4B} sample, $p = 0.25$ versus Control_{4C}, and $p = 0.018$ versus RG₄. We treat these raw tests as descriptive because they do not adjust for stellar mass, rank, or group-scale differences.

The same comparison restricted to BGGs and satellites is also shown in Table 2. For BGGs, the p-values obtained with two-sided Fisher exact test are $p = 0.578$ against Control_{4B}, $p = 0.587$ against Control_{4C} and $p = 0.37$ against RG₄. For satellites, the corresponding p-values are $p = 0.303$ against Control_{4B}, $p = 0.033$ against Control_{4C} and $p = 2.3 \times 10^{-4}$ against RG₄. The satellite deficit is clearest in the raw comparison with RG₄ and is revisited below with adjusted and matched analyses.

Sample		Quenched	Passive	Star-forming
CG ₄		18 (7.3 %)	146 (58.9 %)	84 (33.9 %)
	<i>BGG</i>	6 (9.7 %)	45 (72.6 %)	11 (17.7 %)
	<i>Satellites</i>	12 (6.5 %)	101 (54.3 %)	73 (39.2 %)
Control _{4B}		181 (6.5 %)	1596 (57.1 %)	1019 (36.4 %)
	<i>BGG</i>	58 (8.3 %)	533 (76.4 %)	107 (15.3 %)
	<i>Satellites</i>	122 (5.8 %)	1061 (50.7 %)	911 (43.5 %)
Control _{4C}		191 (6.3 %)	1619 (53.8 %)	1198 (39.8 %)
	<i>BGG</i>	65 (8.7 %)	568 (75.6 %)	118 (15.7 %)
	<i>Satellites</i>	125 (5.5 %)	1050 (46.6 %)	1078 (47.8 %)
RG ₄		5 (2.2 %)	101 (45.1 %)	118 (52.7 %)
	<i>BGG</i>	3 (5.4 %)	38 (67.9 %)	15 (26.8 %)
	<i>Satellites</i>	2 (1.2 %)	63 (37.5 %)	103 (61.3 %)
<i>SDSS selection</i>		634 (1.2 %)	7705 (14.7 %)	44192 (84.1 %)

Table 2. Number of galaxies in each sSFR status for each sample.

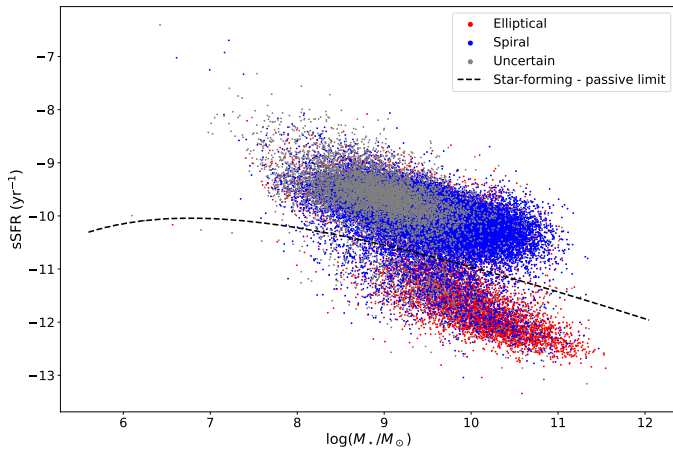


Fig. 2. sSFR vs mass, star-forming/passive limit and Zoo morphologies for the SDSS control sample, excluding quenched galaxies.

2.4. Morphology classification

Morphologies are determined via the Galaxy Zoo citizen-science decision tree, in which each galaxy image receives multiple independent volunteer classifications that are aggregated into debiased vote fractions (Lintott et al. 2011). Those fractions are provided in the zooSpec table of the SDSS database through the `p_el_debiased` and `p_cs_debiased` columns. We classify a galaxy as “elliptical/smooth” if `p_el_debiased` is greater than 0.5, and as “spiral/features” if `p_cs_debiased` is greater than 0.5. All other cases are classified as “uncertain”. These are vote-fraction classes, not a full structural decomposition. In particular, the catalogue used here does not provide a clean E/S0 separation, so references below to elliptical or early-type galaxies should be read as Galaxy Zoo smooth/early-type proxies rather than as a bulge-disc structural classification.

2.5. Statistical strategy

The analyses are organized hierarchically. Raw tables and exact tests provide descriptive baseline comparisons. Fisher or Barnard exact tests are used for two-sample categorical comparisons depending on the table structure, while rank-

sum or bootstrap comparisons are used for distributional and median differences. Spearman and Pearson correlations are used only as diagnostics of monotonic or linear trends; when the coefficients are small, we interpret them as weak even if the p-values are formally small. Holm or Benjamini-Hochberg corrections are applied where multiple related tests are performed. The primary robustness layer consists of mass-, rank-, and environment-adjusted binomial models with group-clustered standard errors, followed by one-to-one matched control comparisons. Colours, AGN-like fractions, H α equivalent widths, magnitude gaps, crossing-time trends, projected phase space, and the pairwise tidal index are secondary diagnostics. The main conclusions below rely on signals that survive the adjusted and matched analyses, not on isolated exploratory tests.

3. Baseline population differences

3.1. Raw morphology fractions

The morphologies obtained for each sample are shown in Table 3. Barnard’s two-sided exact tests were used to compare the fraction of each morphology between CG₄ and each control sample. The resulting p-values are $p = 1.4 \times 10^{-3}$ for ellipticals and $p = 1.2 \times 10^{-3}$ for spirals versus the Control_{4B} sample, $p = 3.3 \times 10^{-3}$ and $p = 0.034$ versus Control_{4C} and $p < 10^{-6}$ and $p < 10^{-6}$ versus RG₄. The galaxies in CG₄ systems, which we identified as cores of richer groups in Tricottet et al. (2025), show a robust raw excess of elliptical/smooth classifications and a deficit of spiral/features classifications. Figure 3 displays sSFR versus stellar mass for galaxies in the SDSS selection and in CG₄.

3.2. Morphology versus sSFR

Among star-forming CG₄ galaxies, 58 are classified as spiral (84.1% after removing galaxies with uncertain morphology) and 11 are classified as elliptical (15.9% after removing galaxies with uncertain morphology). The corresponding numbers are 809 spiral galaxies (88.4%) and 106 elliptical galaxies (11.6%) for Control_{4B}, 851 spiral galaxies (84.3%) and 159 elliptical galaxies (15.7%) for Control_{4C}, and 91 spiral galaxies (91%) and 9 elliptical galaxies (9%) for RG₄. The probabilities that these fractions are drawn from the same distribution, estimated with Barnard two-

Sample	Elliptical/smooth	Spiral/features	Uncertain
CG ₄	124 (50.0 %)	86 (34.7 %)	38 (15.3 %)
Control _{4B}	1106 (39.6 %)	1266 (45.3 %)	420 (15.0 %)
Control _{4C}	1215 (40.4 %)	1249 (41.6 %)	540 (18.0 %)
RG ₄	60 (26.8 %)	129 (57.6 %)	35 (15.6 %)
SDSS selection	10091 (19.2 %)	34715 (66.1 %)	7725 (14.7 %)

Table 3. Number of galaxies in each Galaxy Zoo vote-fraction morphology proxy for each sample.

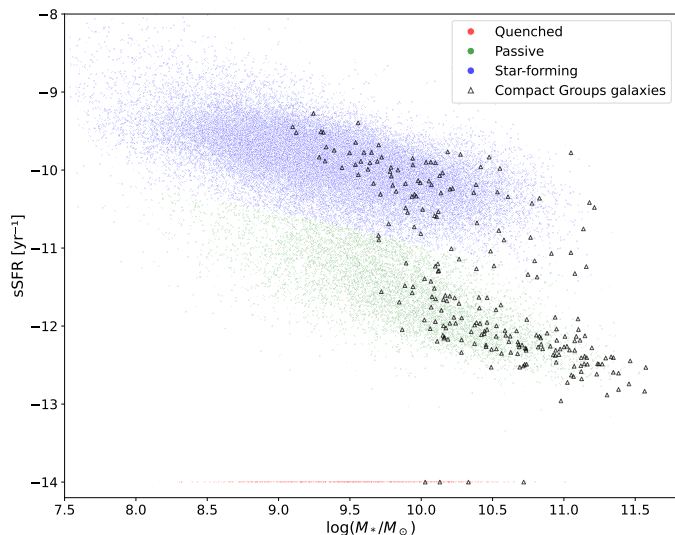


Fig. 3. sSFR and morphologies of galaxies in CG₄ and our SDSS selection. Quenched galaxies are represented at an arbitrary display-only floor of $10^{-14.0} \text{ yr}^{-1}$.

sided exact test, are $p = 0.287$ against Control_{4B}, $p = 0.969$ against Control_{4C}, and $p = 0.186$ against RG₄. We therefore find neither a significant excess nor a deficit of star-forming elliptical galaxies in CG₄ relative to any control sample.

When restricting the comparison to BGGs, we find 36 elliptical BGGs (58.1% of the BGGs) and 17 spiral BGGs (27.4%) in CG₄. The corresponding numbers are 405 elliptical BGGs (58.0%) and 195 spiral BGGs (27.9%) in Control_{4B}, 424 elliptical BGGs (56.5%) and 216 spiral BGGs (28.8%) in Control_{4C}, and 21 elliptical BGGs (37.5%) and 26 spiral BGGs (46.4%) in RG₄. The p-values obtained with two-sided Fisher exact test are 1.00 against Control_{4B}, 0.88 against Control_{4C} and 0.03 against RG₄. Thus, BGG morphologies differ between CG₄ and RG₄: CG₄ BGGs have a higher elliptical fraction than RG₄ BGGs.

3.3. Population medians by sSFR class

Tables 4, 5, and 6 summarize the typical stellar masses, absolute magnitudes, and sSFRs of passive and star-forming galaxies in the different samples, including splits by the sSFR class of the BGG. These medians are descriptive rather than causal: they show that star-formation class is entangled with stellar mass, luminosity, and group role. This motivates the adjusted models in Sect. 5, where stellar mass, rank, and group-scale quantities are treated explicitly.

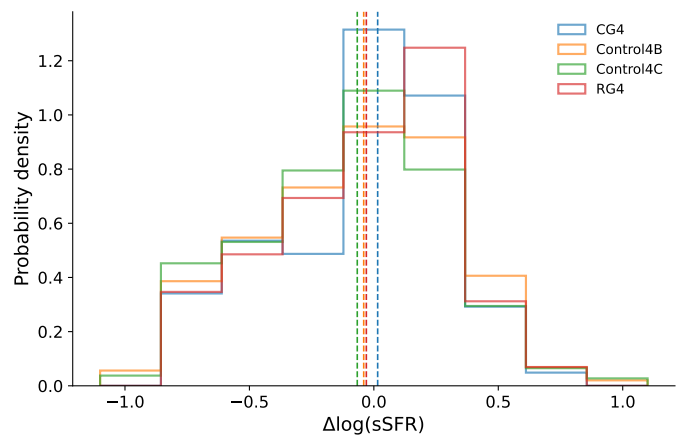


Fig. 4. Main-sequence residual distributions for the star-forming galaxies in each sample.

3.4. Main sequence

We model the star-forming main sequence as an empirical function of stellar mass using polynomial orders 1–4; the polynomial-order comparison is shown in Appendix Fig. A.5. We adopt the quadratic fit as a flexible but still smooth reference relation for residual measurements. The qualitative conclusions below do not rest on interpreting this polynomial as a physical model of the main sequence. We then compute the residual of each star-forming galaxy with respect to this reference sequence and show the resulting distributions in Fig. 4.

Positive residuals mean higher sSFR than the fitted star-forming sequence at fixed stellar mass. For each control sample below, Δ is defined as the median residual of CG₄ star-forming galaxies minus the median residual of the control star-forming galaxies. The quoted bracket is the central 16th–84th percentile bootstrap interval; the p-value is the corresponding two-sided bootstrap sign probability for the median difference.

- The CG₄ residual distribution is marginally different from Control_{4B}: $\Delta = 0.056$, with central 16th–84th percentile interval $[0.027, 0.098]$ and 95% bootstrap interval $[-0.001, 0.147]$, corresponding to $p = 0.058$.
- The CG₄ residual distribution is significantly different from Control_{4C}: $\Delta = 0.082$, with central 16th–84th percentile interval $[0.062, 0.123]$ and 95% bootstrap interval $[0.036, 0.167]$, corresponding to $p = 0.011$.
- The CG₄ residual distribution is not significantly different from RG₄: $\Delta = 0.046$, with central 16th–84th percentile interval $[-0.037, 0.098]$ and 95% bootstrap interval $[-0.086, 0.150]$, corresponding to $p = 0.532$.

The main sequence residual comparison is the diagnostic used for the interpretation. The catalogue also stores

Sample	$\log_{10}(\text{sSFR} [\text{yr}^{-1}])$		M_r		$\log(M_\star/M_\odot)$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	-12.08 ± 0.03	-10.19 ± 0.07	-21.16 ± 0.14	-20.4 ± 0.14	10.64 ± 0.07	10.05 ± 0.05
Control4B	-12.1 ± 0.01	-10.31 ± 0.02	-21.52 ± 0.04	-20.75 ± 0.04	10.79 ± 0.01	10.24 ± 0.02
Control4C	-11.92 ± 0.02	-10.16 ± 0.02	-20.81 ± 0.07	-19.93 ± 0.04	10.47 ± 0.03	9.82 ± 0.03
RG4	-11.91 ± 0.07	-10.16 ± 0.07	-21.11 ± 0.24	-20.32 ± 0.13	10.62 ± 0.08	9.97 ± 0.09

Table 4. Median values of galaxy properties for passive and star-forming populations in each sample. Uncertainties are estimated by bootstrapping (10 000 iterations).

Sample	$\log_{10}(\text{sSFR} [\text{yr}^{-1}])$		M_r		$\log(M_\star/M_\odot)$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	-11.53 ± 0.09	-10.85 ± 0.29	-20.93 ± 0.11	-20.77 ± 0.32	10.46 ± 0.06	10.32 ± 0.12
Control4B	-11.53 ± 0.03	-10.86 ± 0.06	-21.25 ± 0.03	-21.12 ± 0.07	10.61 ± 0.02	10.46 ± 0.04
Control4C	-11.3 ± 0.03	-10.62 ± 0.06	-20.42 ± 0.05	-20.55 ± 0.08	10.22 ± 0.02	10.15 ± 0.06
RG4	-11.14 ± 0.18	-10.58 ± 0.16	-20.69 ± 0.14	-20.74 ± 0.27	10.32 ± 0.07	10.19 ± 0.12

Table 5. Median values of various quantities for galaxies according to the sSFR class of their BGG (BGG included). Uncertainties are estimated by bootstrapping (10 000 iterations).

Sample	$\log_{10}(\text{sSFR} [\text{yr}^{-1}])$		M_r		$\log(M_\star/M_\odot)$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	-11.23 ± 0.16	-10.92 ± 0.48	-20.45 ± 0.13	-20.34 ± 0.18	10.22 ± 0.05	10.09 ± 0.07
Control4B	-11.26 ± 0.06	-10.91 ± 0.13	-20.9 ± 0.03	-20.75 ± 0.06	10.43 ± 0.02	10.3 ± 0.03
Control4C	-10.96 ± 0.04	-10.6 ± 0.07	-19.8 ± 0.04	-20.02 ± 0.09	9.91 ± 0.02	9.89 ± 0.07
RG4	-10.82 ± 0.19	-10.61 ± 0.22	-20.23 ± 0.16	-20.3 ± 0.13	10.09 ± 0.08	10 ± 0.13

Table 6. Median values of various quantities for galaxies according to the sSFR class of their BGG (satellites only). Uncertainties are estimated by bootstrapping (10 000 iterations).

a separate `sSFR_MS_offset` quantity, and the exploratory table below applies a directional rank-sum comparison to that offset. Because this is a different statistic and test, it is used only as a secondary consistency check and not as evidence for an independent main-sequence shift.

A direct rank-sum comparison of the star-forming main-sequence offsets only identifies significant differences between CG₄ and

Control_{4C}, whose median offset is -0.060 with respect to the fitted sequence ($p = 0.03$).

Within CG₄, the star-forming main-sequence offsets of the Embedded, Isolated and Predominant classes remain statistically consistent with one another.

4. Environmental and group-scale diagnostics

4.1. Distance to the BGG

We also compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

Significant monotonic trends are detected in the following cases.

In CG₄, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.24$ for 177 satellites ($p = 1.2 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 4.1 \times 10^{-3}$).

In Control_{4B}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.08$ for 1991 satellites ($p = 3.6 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 2.3 \times 10^{-4}$).

In Control_{4C}, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.25$ for 2143 satellites ($p < 10^{-6}$) and Pearson coefficient $r = 0.20$ ($p < 10^{-6}$).

In Control_{4C}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.10$ for 2143 satellites ($p = 4.6 \times 10^{-6}$) and Pearson coefficient $r = 0.09$ ($p = 1.8 \times 10^{-5}$).

The supporting distance plots are shown in Appendix Figs. A.6 and A.7.

4.2. Dominance

We split groups into dominated systems, where the BGG contributes more than 60% of the group luminosity, and non-dominated systems. This split does not produce a strong compact-group signal by itself: CG₄ shows no significant dominated/non-dominated difference after the Benjamini-Hochberg false-discovery-rate (FDR) correction. Here p_{adj} denotes the FDR-adjusted p-value, and Cliff's δ is a rank-based effect size ranging from -1 to 1.

Across the control and reference samples, the clearest binary separation is instead in group luminosity. Dominated systems have lower L_{group} than their non-dominated counterparts: Control_{4B} gives 9.1×10^{10} versus $1.28 \times 10^{11} L_\odot$ ($p_{\text{adj}} < 10^{-6}$, $\delta = -0.55$); Control_{4C} gives 8.49×10^{10} versus $9.54 \times 10^{10} L_\odot$ ($p_{\text{adj}} < 10^{-6}$, $\delta = -0.25$); and RG₄ gives 7.09×10^{10} versus $9.82 \times 10^{10} L_\odot$ ($p_{\text{adj}} = 7.8 \times 10^{-5}$, $\delta = -0.59$). The negative δ values mean that dominated groups tend to occupy the lower-luminosity side of each comparison.

The spiral-fraction correlations show a broader environmental pattern rather than a clean dominated/non-dominated dichotomy. The most coherent set appears in Control_{4C}, where S_{frac} increases with $\langle R_{ij} \rangle$ in both dominated and non-dominated groups ($\rho_S = 0.38$, $p < 10^{-6}$ and $\rho_S = 0.35$, $p < 10^{-6}$), while weaker negative trends are found with σ_v and L_{group} . Control_{4B} contributes only weaker non-dominated-group correlations. In CG₄, the significant trends occur in small subsets, with S_{frac} decreasing with L_{group} among dominated groups ($\rho_S = -0.50$, 36 groups) and with $\Delta V_{\text{BGG}}/\sigma_v$ among non-dominated groups ($\rho_S = -0.53$, 26 groups); we therefore treat these as supporting diagnostics rather than the main dominance result. The full distributional and correlation diagnostics are shown in Appendix Figs. A.8 and A.9.

4.3. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, significant effects only appear in

Control_{4B} dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The corresponding group-level odds ratio is 1.842 ($p = 0.01$), while the satellite-level clustered fit gives an odds ratio of 1.842 ($p = 0.02$);

Control_{4C} dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.50 for elliptical BGGs. The corresponding group-level odds ratio is 1.840 ($p = 2.1 \times 10^{-5}$), while the satellite-level clustered fit gives an odds ratio of 1.840 ($p = 2.7 \times 10^{-5}$).

4.4. Crossing time

The crossing-time diagnostic mainly separates systems by dynamical scale rather than by luminosity alone. Shorter t_{cross} systems have larger virial mass-to-light ratios and virial masses: the correlations with $\log(M_{\text{VT}}/L_r)$ and $\log(M_{\text{VT}})$ are $\rho_S = -0.39$ ($p < 10^{-6}$) and $\rho_S = -0.33$ ($p < 10^{-6}$), respectively, over 816 groups. By contrast, the relation with total group luminosity is formally significant but very weak ($\rho_S = 0.08$, $p = 0.03$). We therefore read t_{cross} as a dynamical compactness indicator tied primarily to virial mass and mass-to-light ratio, with little independent leverage from L_{group} . Appendix Figs. A.10–A.12 show the corresponding projected and three-dimensional trends.

4.5. Optical colours

We derive the $(u - r)$, $(u - g)$, $(g - r)$ and $(r - i)$ colours from the matched SDSS photometry. The retained matches include 110 of 248 galaxies in CG₄, 931 of 2792 galaxies in Control_{4B}, 1478 of 3004 galaxies in Control_{4C}, 102 of 224 galaxies in RG₄. Stellar mass is a confounding variable in colour comparisons (Alpaslan et al. 2015); we therefore model each colour jointly with $\log(M_{\star}/M_{\odot})$ rather than comparing unadjusted colour distributions.

The colour-matched subset is not representative of the full galaxy catalogues: matched galaxies are systematically less massive, more actively star-forming, and more frequently satellites than unmatched galaxies. The fractions with all four colours are 44.4% in CG₄, 33.3% in Control_{4B}, 49.2% in Control_{4C}, 45.5% in RG₄. The colour analysis is

therefore useful as a secondary diagnostic, but it cannot carry the main interpretation by itself.

The raw catalogue colour–mass fits indicate that CG₄ does not simply follow the same colour–mass relation as every comparison sample: common slopes are rejected in 4 of the four colours, with the significant directional slope differences concentrated in the bluer colours and dependent on the adopted comparison catalogue. In the satellite-only comparison, the fitted offsets and slopes likewise depend on the control definition. These tests motivate the robustness analysis rather than establishing a stand-alone compact-group colour effect.

We therefore refit the satellite colours with cluster-robust standard errors, using catalogue-qualified group identifiers. In the pooled comparison adjusted only for stellar mass and redshift, none of the four CG₄ coefficients is significantly negative after Holm correction. After additionally controlling for morphology and sSFR class, negative coefficients appear in $(u - r)$ ($\Delta = -0.127$ mag, Holm-adjusted $p = 0.018$) and $(u - g)$ ($\Delta = -0.101$ mag, Holm-adjusted $p = 0.027$). However, these adjusted coefficients are control-sample dependent, and they do not persist within the sSFR or morphology splits. The data therefore do not localize the colour effect to the star-forming spiral population expected for a simple interaction-triggered star-formation scenario.

Residual colour trends with projected BGG-centric distance are consistent with stronger processing near the group centre and a less processed outer population, but these trends use the same biased colour subset and are treated as exploratory. The group-level BGG–satellite colour concordance is similarly secondary: redder BGGs tend to have redder mean satellite populations in the combined group sample, but the CG₄ slope is not a robustly distinct population-level signature. Detailed colour–mass, satellite, radial, BGG–satellite, and domination diagnostics are shown in Appendix Figs. A.13–A.17. Figure 5 summarizes the model and control-sample dependence that motivates our interpretation: the present data do not establish a robust global blue or red offset for CG₄ satellites.

5. Robustness: what drives the compact-group differences?

The previous sections establish that CG₄ galaxies differ from the control catalogues in their population mix, especially in morphology and in the star-forming fraction of satellites. We now ask whether these differences survive after controlling for stellar mass, rank and available group-scale quantities, and whether they are better interpreted as compact-group-specific evolution, preprocessing in dense halo cores, or catalogue-selection effects.

5.1. Mass-, rank- and environment-adjusted galaxy properties

We fit binomial models to the combined galaxy catalogues, standardizing the continuous predictors and using group-clustered standard errors. The common adjustment set includes stellar mass, redshift, satellite/BGG rank, group luminosity and velocity dispersion when sufficiently complete.

For passive status, the CG₄ odds ratio is 1.20, with 95% confidence interval [0.88, 1.63] and $p_{\text{Holm}} = 0.485$.

For passive satellite status, the CG₄ odds ratio is 1.31, with 95% confidence interval [0.95, 1.80] and $p_{\text{Holm}} = 0.390$.

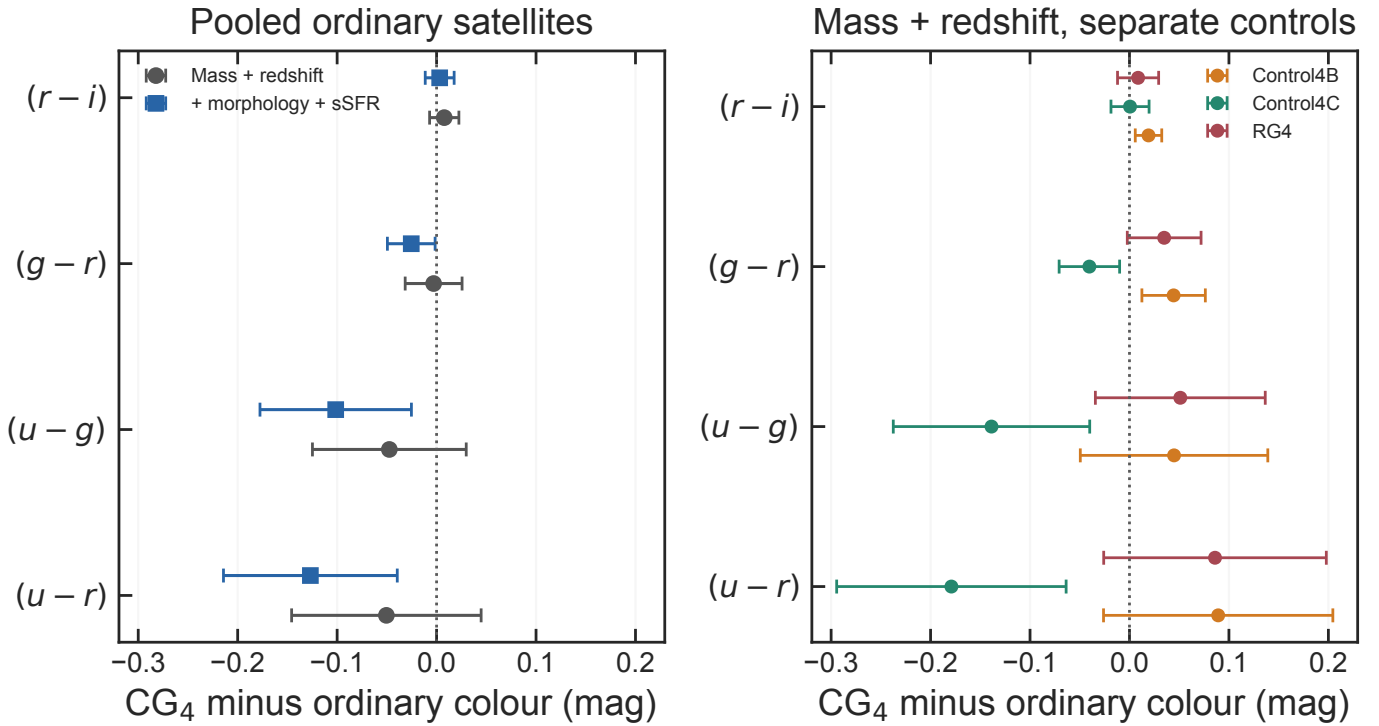


Fig. 5. Robustness of the satellite colour coefficients. Left: pooled ordinary-group comparison before and after controlling for morphology and sSFR class. Right: mass- and redshift-adjusted comparisons against each control catalogue separately. Points show the CG_4 minus ordinary colour coefficient and bars show 95% confidence intervals; negative values indicate bluer CG_4 satellites.

For star-forming satellite status, the CG_4 odds ratio is 0.76, with 95% confidence interval [0.56, 1.05] and $p_{\text{Holm}} = 0.390$.

For elliptical morphology, the CG_4 odds ratio is 1.62, with 95% confidence interval [1.21, 2.16] and $p_{\text{Holm}} = 0.007$.

For elliptical satellite morphology, the CG_4 odds ratio is 1.84, with 95% confidence interval [1.34, 2.53] and $p_{\text{Holm}} = 0.001$.

For elliptical BGG morphology, the CG_4 odds ratio is 0.90, with 95% confidence interval [0.46, 1.74] and $p_{\text{Holm}} = 0.751$.

For spiral morphology, the CG_4 odds ratio is 0.62, with 95% confidence interval [0.46, 0.83] and $p_{\text{Holm}} = 0.007$.

For spiral satellite morphology, the CG_4 odds ratio is 0.54, with 95% confidence interval [0.39, 0.75] and $p_{\text{Holm}} = 0.001$.

Compact-group membership therefore remains independently associated with morphology after the available adjustments: CG_4 galaxies have a higher probability of being elliptical and a lower probability of being spiral. In contrast, the adjusted passive and star-forming satellite coefficients are weaker and do not provide evidence for an instantaneous star-formation difference that remains significant after correction.

The satellite-only fits show that this adjusted morphology signal is carried primarily by the satellite population rather than by a separately significant BGG-only coefficient.

5.2. Matched-control comparison

We matched 234 of the 248 available CG_4 galaxies one-to-one to 234 ordinary-group galaxies without replacement.

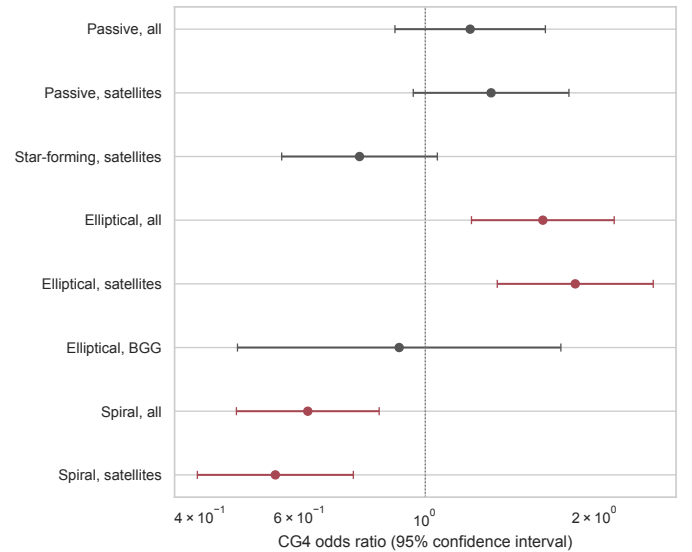


Fig. 6. Adjusted CG_4 odds ratios. Bars show 95% confidence intervals; the vertical line marks equal odds.

The remaining 14 CG_4 galaxies were excluded because they lay outside the common propensity-score support or exceeded the adopted caliper. The matching required common propensity-score support, a 0.2-standard-deviation caliper, and exact rank strata. The propensity model uses logMstar, z numeric, rank, log group luminosity, velocity dispersion. Projected distance is retained as a phase-space and interaction diagnostic rather than matched away because it partly defines compactness and has limited overlap. The largest

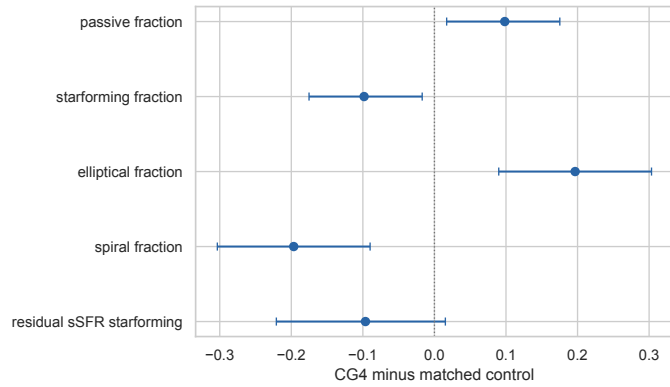


Fig. 7. CG₄-minus-control effects after one-to-one matching. Bars show 95% bootstrap confidence intervals.

absolute standardized mean difference changed from 0.28 before matching to 0.14 afterwards. The remaining maximum imbalance is slightly above the common 0.1 heuristic for excellent balance, so we treat the matched comparison as a robustness check rather than as a fully balanced pseudo-experiment.

The matched CG₄-minus-control difference in passive fraction is 0.098, with 95% bootstrap interval [0.017, 0.175] and $p_{\text{Holm}} = 0.054$.

This is suggestive but not formally significant after correction, so we treat the passive-fraction excess as borderline.

The corresponding CG₄-minus-control difference in star-forming fraction is -0.098 , with 95% bootstrap interval $[-0.175, -0.017]$ and $p_{\text{Holm}} = 0.054$.

The matched CG₄-minus-control difference in elliptical fraction is 0.197, with 95% bootstrap interval [0.090, 0.303] and $p_{\text{Holm}} < 10^{-6}$.

The matched CG₄-minus-control difference in spiral fraction is -0.197 , with 95% bootstrap interval $[-0.303, -0.090]$ and $p_{\text{Holm}} < 10^{-6}$.

Among matched star-forming galaxies, the residual-sSFR difference is -0.096 , with 95% bootstrap interval $[-0.221, 0.015]$ and $p = 0.084$.

The main interpretation does not rely on the exact GMM division of the sSFR plane. Repeating the key satellite comparison with a fixed threshold, $\log_{10}(\text{sSFR}/\text{yr}^{-1}) = -11.0$, gives CG₄ and RG₄ satellite star-forming fractions of 40.7% and 60.2%, respectively, a CG₄-RG₄ difference of -19.6 percentage points ($p = 3.6 \times 10^{-4}$). The qualitative hierarchy is unchanged: the morphology result is independent of the sSFR threshold, whereas the passive/star-forming contrast remains more sensitive to modelling choices.

The matched comparison is therefore consistent with the adjusted models: the most robust compact-group signal is morphological, while the passive and star-forming-fraction signals are more model-dependent.

5.3. Galaxy Zoo crowding robustness

Galaxy Zoo morphologies can in principle be affected by close projected neighbours. We therefore repeat a targeted morphology check after excluding galaxies whose nearest same-sample group member lies within 55 arcsec. This removes 48 of 248 CG₄ galaxies and 384 of 3004 Control_{4C} galaxies. After this exclusion, the CG₄ elliptical/smooth

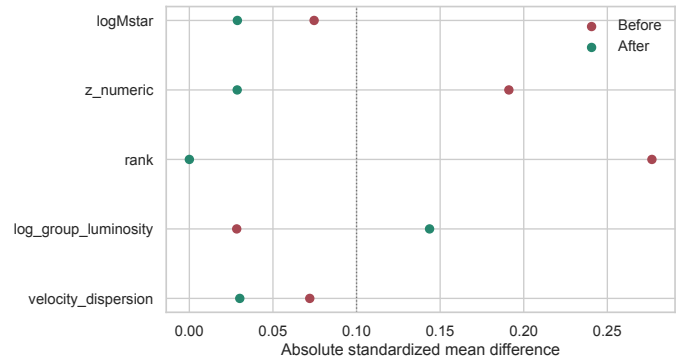


Fig. 8. Absolute standardized mean differences before and after matching.

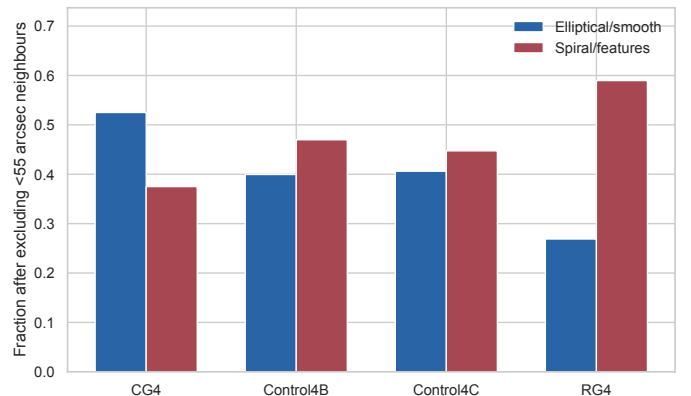


Fig. 9. Galaxy Zoo morphology fractions after excluding galaxies with nearest same-sample group neighbours within 55 arcsec.

and spiral/features fractions are 52.5% and 37.5%, compared with 40.6% and 44.7% for Control_{4C}.

The close-neighbour-excluded exact tests against the most compact-core-like control give $p = 0.001$ for elliptical/smooth morphology and $p = 0.047$ for spiral/features morphology.

In the adjusted crowding model, the CG₄ elliptical/smooth odds ratio is 1.54 (95% CI 1.14–2.07; $p = 0.004$), while the close-neighbour flag has odds ratio 1.93 ($p = 2.9 \times 10^{-6}$). This does not prove that image blending is irrelevant, but it shows that the morphology signal is not solely a product of the closest projected pairs.

An independent SDSS concentration-index check is deferred because the current processed catalogue lacks the r-band Petrosian radii needed for R_{90}/R_{50} .

5.4. Satellite segregation around the brightest group galaxy

The previous sections show global differences between CG₄ and RG₄ satellites. We now ask whether these differences are concentrated near the BGG, as expected for accelerated evolution in the dense inner group, or whether they persist across the satellite population. The analysis uses 177 CG₄ and 166 RG₄ satellites, excludes rank-1 galaxies, ranks satellites by projected BGG-centric distance, and measures passive and early-type fractions with group-cluster bootstrap uncertainties. When available, we also use $|\Delta v_{\text{BGG}}|/\sigma_v$, with $\Delta v_{\text{BGG}} = c(z_{\text{gal}} - z_{\text{BGG}})/(1 + z_{\text{group}})$, as a projected phase-space diagnostic. Similar projected phase-space co-

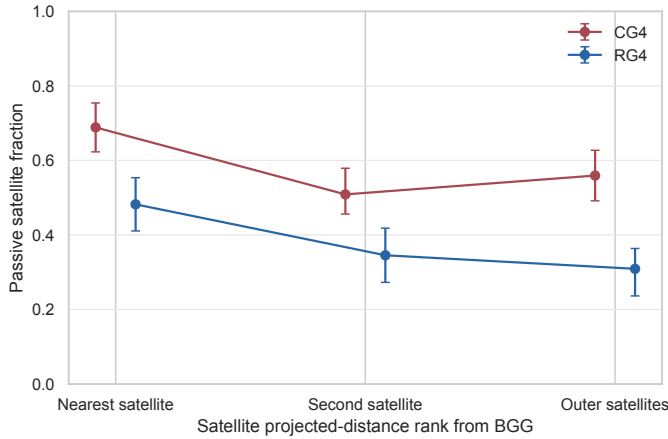


Fig. 10. Passive satellite fraction as a function of projected-distance rank from the BGG. Points show CG₄ and RG₄ fractions; bars show group-cluster bootstrap 16th–84th percentile intervals.

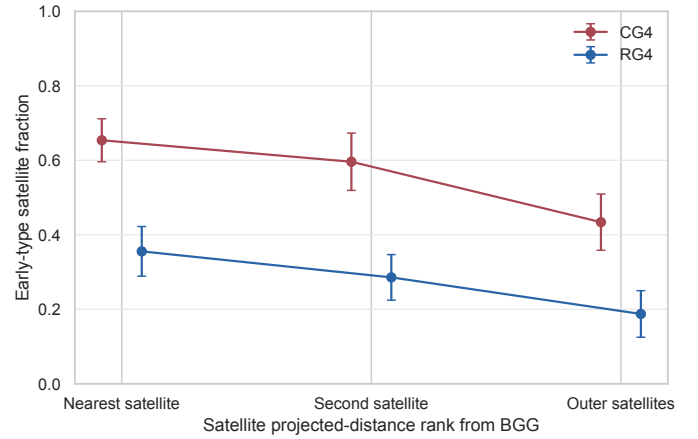


Fig. 11. Early-type satellite fraction as a function of projected-distance rank from the BGG. Because the catalogue separates ellipticals, spirals, and uncertain morphologies but not S0 galaxies, early type is equivalent to elliptical in this diagnostic.

ordinates are often used as observational proxies for satellite infall state or quenching time, but only statistically (Mahajan et al. 2011; Oman & Hudson 2016; Wetzel et al. 2013).

In the innermost satellite-distance bin, the passive fractions are 68.9% for CG₄ and 48.2% for RG₄, a CG₄–RG₄ difference of 20.6 percentage points. The corresponding early-type fractions are 65.4% and 35.6%, with a difference of 29.8 percentage points.

The passive-fraction contrast is similar between the inner and outer distance ranks, so the data do not isolate the CG₄ offset to the immediate BGG environment.

After adjusting for stellar mass, redshift, distance bin, and available group-scale controls, the passive-status model gives a CG₄ odds ratio of 1.27 (95% CI 0.47–3.46; $p = 0.640$).

The analogous early-type model gives a CG₄ odds ratio of 2.57 ($p = 0.058$).

A nearest-neighbour mass–redshift matched check uses 128 CG₄–RG₄ satellite pairs. The matched passive-fraction difference is 9.4 percentage points, and the matched early-type difference is 19.6 percentage points.

Projected phase-space coordinates should not be read as orbital reconstructions for individual galaxies. In compact groups the member counts are small, velocity dispersions are noisy, and projection mixes recently accreted and long-resident satellites. We therefore treat these bins as an observational diagnostic of where the CG₄–RG₄ population contrast appears, rather than as a direct dynamical clock.

5.5. Magnitude gaps and fossil-like assembly

We define $\Delta m_{12} = M_{r,2} - M_{r,1}$ and $\Delta m_{14} = M_{r,4} - M_{r,1}$, so that a more dominant BGG has a positive, larger gap.

For Δm_{12} , the CG₄ and control medians are 1.17 and 1.09 mag, respectively ($p_{\text{Holm}} = 0.705$).

For Δm_{14} , the CG₄ and control medians are 2.56 and 2.59 mag, respectively ($p_{\text{Holm}} = 0.705$).

The compact-group morphology signal is therefore not accompanied by a significant excess in these simple fossil-like magnitude-gap indicators.

The correlation between Δm_{12} and passive satellite fraction is $\rho_S = -0.15$ ($p_{\text{Holm}} < 10^{-6}$).

The supporting magnitude-gap distributions are shown in Appendix Figs. A.19 and A.20.

5.6. Recently quenched candidates

The catalogues do not contain both D_n4000 and $H\delta_A$, so a post-starburst classification is not possible. We therefore perform only a limited $H\alpha$ -equivalent-width comparison, using the SDSS convention in which negative equivalent width denotes emission. We classify strong $H\alpha$ emission using $\text{h_alpha_eqw} \leq -3$ Angstrom.

The strong-emission fraction is 32.3% in CG₄, 41.0% in Control4B, 45.7% in Control4C, 56.9% in RG₄.

Relative to Control4B, the CG₄ strong-emission fraction differs by -0.087 ($p_{\text{Holm}} = 0.012$).

Relative to Control4C, the CG₄ strong-emission fraction differs by -0.135 ($p_{\text{Holm}} = 2.4 \times 10^{-4}$).

Relative to RG₄, the CG₄ strong-emission fraction differs by -0.246 ($p_{\text{Holm}} = 1.5 \times 10^{-6}$).

These limited $H\alpha$ results indicate that CG₄ galaxies have a lower strong-emission fraction than in each control sample in the current catalogue, but they do not provide a direct post-starburst or recently-quenched classification.

The $H\alpha$ equivalent-width distributions are shown in Appendix Fig. A.21.

5.7. Emission-line and AGN-like populations

The complementary emission-line analysis classifies BPT-star-forming, composite and AGN-like populations for galaxies with the required diagnostic line ratios, using the Kauffmann et al. (2003) and Kewley et al. (2001) demarcations described in Sect. 2.2. BPT-unclassified galaxies are excluded from the AGN-like fraction denominator rather than treated as confirmed non-AGN. Among BPT-classified CG₄ galaxies, the AGN-like fraction is 38.0%.

After mass and environment adjustment, the CG₄ AGN-like odds ratio is 1.14 ($p = 0.451$).

Thus, AGN-like activity does not provide a robust independent compact-group discriminator in this analysis.

The AGN-like fractions are shown in Appendix Fig. A.22.

5.8. Pairwise tidal index

The pairwise analysis reconstructs projected separations and velocity differences for 6268 galaxies and tests whether adding $\sum_j M_{\star,j}/R_{ij}^3$ changes the adjusted CG₄ coefficient.

For passive status, the baseline CG₄ odds ratio is 1.20 ($p = 0.227$), while the model including the tidal index gives a CG₄ odds ratio of 0.80 ($p = 0.170$). The tidal-index term itself has odds ratio 1.65 ($p < 10^{-6}$).

For elliptical morphology, the baseline CG₄ odds ratio is 1.65 ($p = 4.1 \times 10^{-4}$), whereas the model including the tidal index gives a CG₄ odds ratio of 1.15 ($p = 0.346$). The tidal-index term has odds ratio 1.56 ($p < 10^{-6}$). This suggests that the reconstructed local tidal-density proxy absorbs much of the unadjusted CG₄ morphology coefficient, although the result remains an exploratory projected-pair diagnostic.

5.9. Selection diagnostics

The fractions with complete measurements for the major derived quantities are shown in Appendix Fig. A.23. The colour-matched versus unmatched comparisons quantify shifts in stellar mass, redshift, rank, star-formation class and morphology.

The median nearest-neighbour angular separation is 100.9 arcsec in CG₄ and 209.0 arcsec in the controls; the fractions below the 55-arcsec SDSS fibre-collision scale are 19.4% and 9.3%, respectively.

No major availability difference meets both criteria of corrected significance and a difference exceeding 10 percentage points.

The group-scale availability row is based on the group covariates available for this catalogue: radius scale, velocity dispersion, group luminosity, and BGG luminosity dominance. A harmonized group-mass estimate is not available in these tables, so models that request it omit that covariate.

The colour-selection mass audit is shown in Appendix Fig. A.24.

6. Discussion: drivers of structural evolution in compact groups

6.1. Local projected tidal density and morphology signal

The most robust residual difference between CG₄ and the regular-group controls is morphological. CG₄ galaxies are more likely to be classified as elliptical/smooth and less likely to be classified as spiral/features, and this result persists in both the adjusted binomial models and the one-to-one matched comparison. The result hierarchy therefore points to accumulated structural transformation more strongly than to a present-day star-formation offset.

The pairwise tidal-index experiment strengthens this interpretation while also narrowing it. For elliptical/smooth morphology, the baseline CG₄ odds ratio is 1.65 ($p = 4.1 \times 10^{-4}$). After adding the projected pairwise tidal-index proxy, the CG₄ odds ratio decreases to 1.15 ($p = 0.346$), while the tidal-index term itself has odds ratio 1.56 ($p <$

10^{-6}). This behaviour is illustrated in Fig. 12 and is expected if compact groups are selecting galaxies in locally dense projected configurations where repeated close companions, rather than the catalogue label alone, are linked to the smooth/early-type-like population. Because the proxy is reconstructed from projected separations and catalogue masses, it should be read as a local projected tidal-density diagnostic, not as a direct measurement of the full time-dependent tidal field; line-of-sight alignments can inflate such projected density measures. There is an unavoidable element of circularity in this test: compact groups are partly selected for high projected density, and any R_{ij}^{-3} -weighted quantity is therefore related to the selection itself. Its value here is not to prove an independent causal mechanism, but to translate the categorical compact-group label into a continuous immediate-neighbour density scale. The comparison with Control_{4C} is important in this respect, because that control is also built from BGG-centred close projected companions; the tidal-index result should therefore be read as evidence that the most extreme local projected configurations, rather than group membership alone, are associated with the smooth-morphology signal.

6.2. Smooth morphologies, S0 degeneracy, and transition galaxies

The Galaxy Zoo morphology variables used here separate smooth, featureless systems from galaxies with visible spiral structure or other features. They do not cleanly distinguish classical ellipticals from S0 galaxies, faded disks, or other smooth disk-dominated systems. The enhanced CG₄ elliptical/smooth fraction should therefore be interpreted as a shift toward smooth or early-type-like appearances, not as proof that compact groups specifically build classical elliptical galaxies.

This caveat is important when comparing with the recent S-PLUS structural studies of compact-group galaxies by Montaguth et al. (2023, 2025, 2026b,a). Those studies identify transition galaxies and trends in the R_e -Sérsic-index and mass-size planes, with especially strong signals in non-isolated compact groups and host structures. Their transition population can plausibly overlap with the smooth class in our catalogue: systems with faded disks, increasing central concentration, or S0-like structure may be counted as smooth here even if they are not classical ellipticals. The two approaches are therefore complementary. Their structural measurements describe the likely physical form of the transformation, while our regular-group controls show that the strongest residual compact-group signal is the accumulated smooth/featured morphology balance.

The targeted close-neighbour check adds one guardrail to this interpretation: the morphology contrast remains in the same direction after excluding galaxies with nearest same-sample group neighbours within 55 arcsec and in models that include a close-neighbour flag. This is not a substitute for independent structural photometry, but it argues against the signal being only the result of the closest projected pairs affecting Galaxy Zoo classifications.

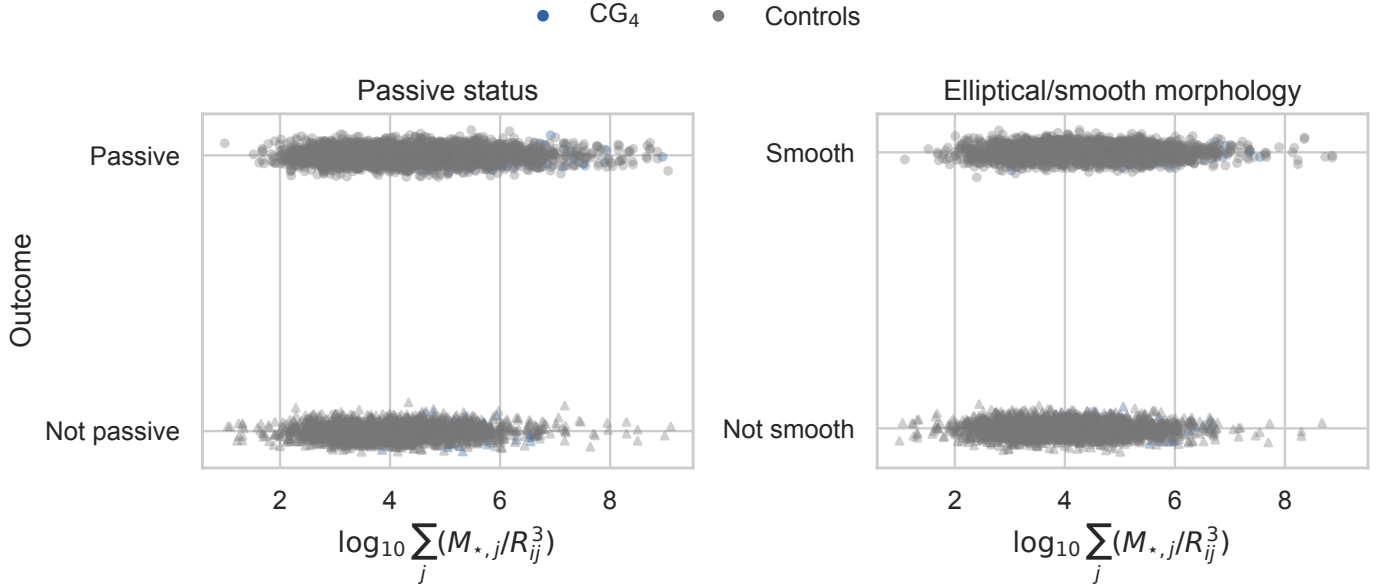


Fig. 12. Passive status and elliptical/smooth morphology as functions of the reconstructed pairwise tidal index. Colours distinguish CG₄ from control galaxies.

Diagnostic	Baseline CG ₄ signal	Survives adjusted/ matched analysis?	Interpretation
Elliptical/smooth fraction	Enhanced in CG ₄	Yes	Robust accumulated morphology signal.
Spiral/features fraction	Reduced in CG ₄	Yes	Reduced late-type vote-fraction population.
Close-neighbour morphology check	Same sign after excluding < 55 arcsec neighbours	Yes	Not solely a closest-pair crowding artefact.
Satellite star-forming fraction	Reduced, especially versus RG ₄	Weak or borderline	Partly explained by mass, rank, environment, or preprocessing.
Star-forming main-sequence residual	Weak and control-dependent	No robust offset	No strong instantaneous SFR enhancement or suppression among remaining star-forming galaxies.
Satellite colours	Colour-slope and residual hints	Control-sensitive	Exploratory; not a simple uniform reddening or blueing of CG ₄ satellites.
AGN-like fraction	No robust independent signal	No	Not an AGN-driven explanation.
Magnitude gaps	No significant excess	No	Not fossil-like under the simple gap diagnostics used here.
Pairwise tidal index	Local projected density proxy is informative	Absorbs much of the morphology coefficient	Local tidal density likely contributes to the morphology signal.

Table 7. Result hierarchy. Baseline tests describe where CG₄ differs from the controls; the main interpretation is based on diagnostics that remain after adjusted or matched comparisons.

6.3. Decoupling between morphology and instantaneous star formation

The star-formation diagnostics do not follow the morphology signal one-to-one. CG₄ satellites have a lower raw star-forming fraction, especially relative to RG₄, but the adjusted and matched evidence is weaker than for morphology. Among galaxies that remain star-forming, the main-sequence residuals do not show a robust compact-group offset. The colour analysis is also control-sensitive and based on a biased colour-matched subset. At fixed morphology

and sSFR class, the tentative bluer CG₄ satellite coefficients are best read as exploratory hints about residual population mix or dust/stellar-population differences, not as evidence for a uniform interaction-triggered starburst. AGN-like fractions, magnitude gaps, and the limited H α diagnostic do not identify a simple alternative driver.

Spectroscopic incompleteness may also act in the same direction. The fraction of galaxies with a nearest neighbour below the 55-arcsec SDSS fibre-collision scale is 19.4% in CG₄, compared with 9.3% in the controls. This affects

spectral quantities most directly and can underrepresent short-lived starbursts, post-starburst phases, or extreme emission-line objects in the closest compact-group pairs. It does not remove the morphology result, which is based on imaging vote fractions and is checked against the closest-pair exclusion above, but it reinforces the need to treat the weaker sSFR and emission-line conclusions conservatively.

This decoupling suggests that the dominant observable difference is not an instantaneous quenching event caught at the present epoch. A galaxy can lose visible spiral structure, build a smoother disk or bulge-dominated profile, or fade into an S0-like state on timescales that do not require a simultaneously large sSFR residual in the surviving star-forming subset. The present data therefore favour a cumulative morphology or structure signal over a single ongoing star-formation process.

6.4. Compact groups as dense cores and preprocessing sites

The comparison samples were designed to ask whether compact groups remain special relative to ordinary group cores, not merely relative to the field. The answer is mixed but physically useful. CG₄ galaxies retain a robust morphology difference, yet much of the interpretation points toward local density, dense cores, and preprocessing rather than toward an isolated compact-group channel. This is consistent with the construction result that many compact groups are embedded in or associated with larger structures, and with the stronger structural signatures reported for non-isolated compact groups in recent S-PLUS work.

The most conservative interpretation is that compact groups identify the high-density, high-interaction tail of group environments. Some systems may be genuinely compact and dynamically evolved, while others may be dense cores or substructures inside larger haloes. In both cases, the relevant physical quantity is likely the accumulated local interaction history. Future simulations and direct structural measurements, including R_e , Sérsic index, transition-galaxy classifications, and mass-size residuals, are needed to test whether the signal isolated here is unique to compact configurations or is the high-density limit of more general group preprocessing.

7. Conclusion

We have tested whether galaxies in CG₄ compact groups remain special after removing split systems and comparing them with three regular-group analogues: RG₄, Control_{4B}, and Control_{4C}. The comparison was designed to separate compact configurations from ordinary four-member groups, luminous group cores, and projected compact cores around BGGs.

The main result is morphological. CG₄ galaxies have an enhanced elliptical/smooth Galaxy Zoo fraction and a reduced spiral/features fraction. This signal survives the mass-, redshift-, rank-, and group-adjusted binomial models and is also recovered in the one-to-one matched-control analysis.

The adjusted model hierarchy indicates that this morphology signal is primarily a satellite-population signal rather than a separately significant BGG-only effect.

It also remains after a targeted close-neighbour exclusion, so it is not solely a closest-pair Galaxy Zoo crowding artefact.

The star-formation result is weaker. CG₄ satellites have a lower raw star-forming fraction, especially relative to RG₄, but the adjusted and matched evidence is weaker or borderline. Among galaxies that remain star-forming, CG₄ galaxies do not show a robust offset from the fitted star-forming main sequence.

The secondary diagnostics do not identify a simple alternative driver. The colour results are control-sensitive and based on a biased colour-matched subset. AGN-like fractions and magnitude gaps do not explain the morphology signal, the H α diagnostic is limited by missing age-sensitive spectral indices, and projected phase-space trends do not localize the passive difference uniquely to the nearest satellites.

We therefore interpret CG₄ galaxies as a more morphologically evolved population, consistent with accumulated transformation in very dense environments, dense group cores, and/or preprocessing. The pairwise tidal-index experiment suggests that local projected tidal density accounts for part of the compact-group morphology coefficient, so the result should not be read as evidence for a uniquely isolated compact-group quenching mechanism.

Future work should connect this regular-group control framework directly to structural measurements, including R_e , Sérsic index, transition-galaxy classifications, mass-size residuals, and better spectroscopic post-starburst diagnostics. A simple SDSS concentration check should also be added after enriching the catalogue with r-band Petrosian radii. Those measurements would test whether the morphology signal isolated here is the same cumulative structural transformation highlighted by recent S-PLUS studies.

Data availability

The SDSS photometric, spectroscopic, Galaxy Zoo, stellar-mass, and star-formation quantities used in this work are publicly available through the SDSS archive. The group catalogues used for the sample construction are those of Tempel et al. (2017), Lim et al. (2017), and Díaz-Giménez et al. (2018), as described in Sect. 1 and Sect. 2. Derived catalogues, matching products, and analysis scripts will be made available in a public repository before publication.

Code availability

The analysis was performed with custom Python scripts for catalogue matching, sSFR classification, adjusted and matched comparisons, and figure generation. The code will be archived in a public GitHub or Zenodo repository before publication.

Appendix A: Supplementary diagnostic figures

This appendix collects the secondary diagnostic figures referenced in the main text.

A.1. *sSFR* and main-sequence diagnostics

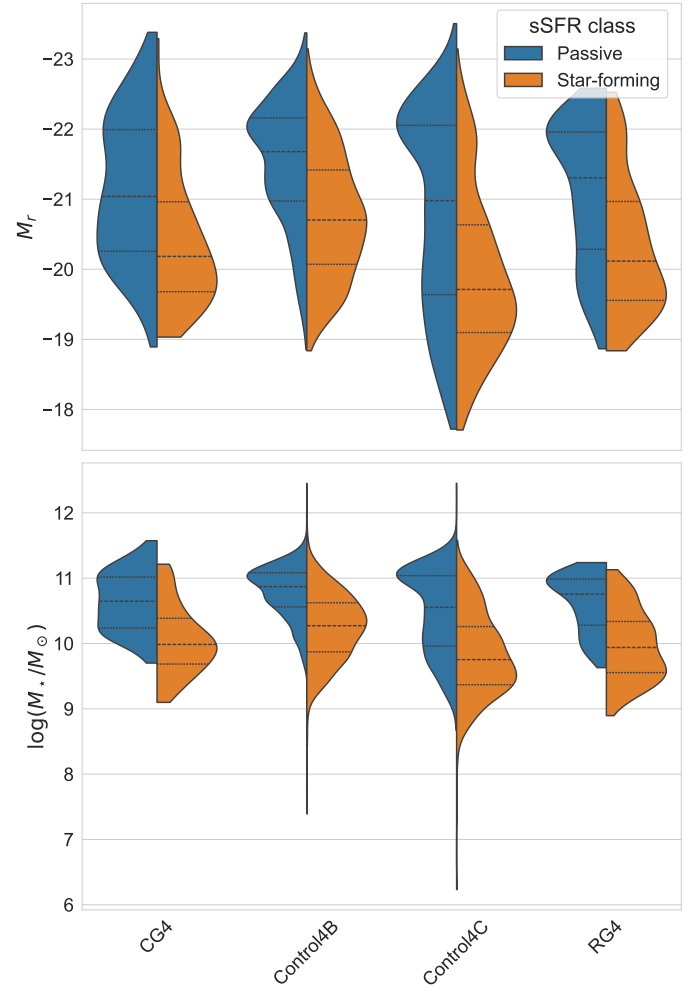


Fig. A.2. Distributions of various quantities for galaxies according to their sSFR class.

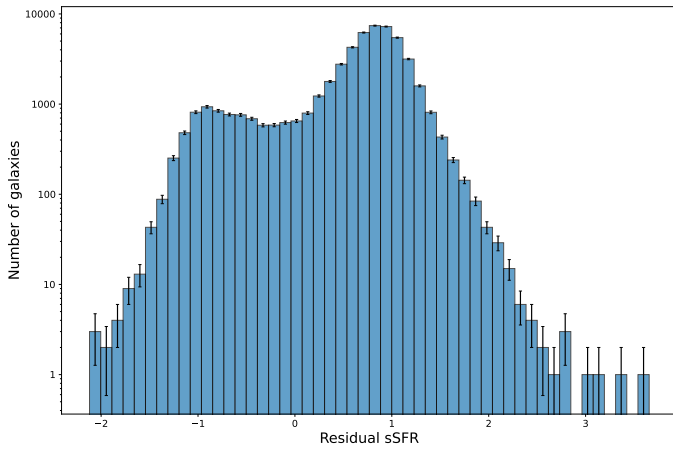


Fig. A.1. Distribution of residual sSFR offsets from the adopted star-forming main sequence.

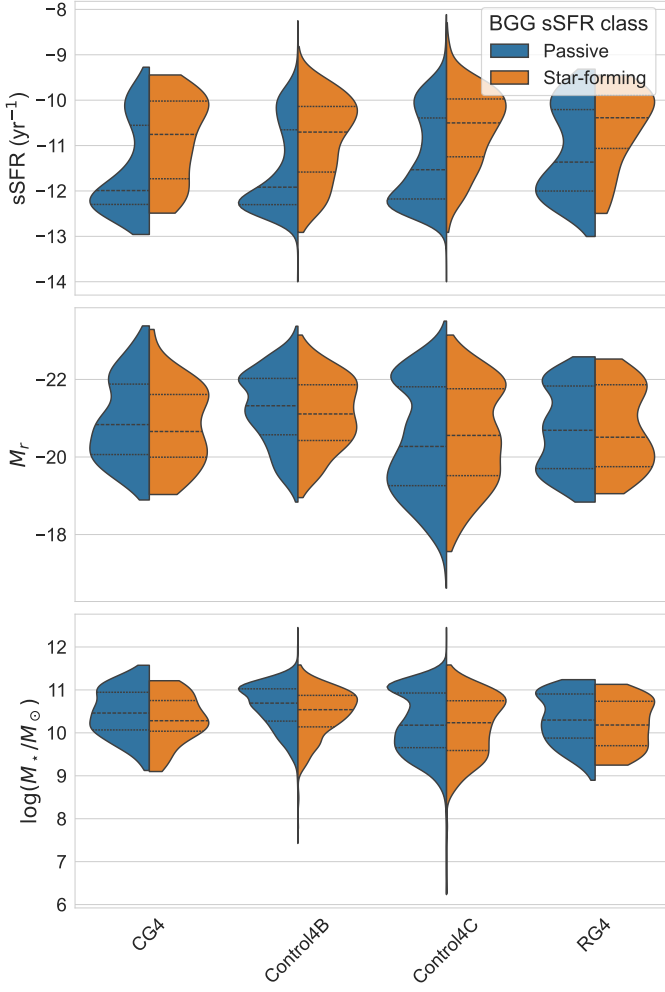


Fig. A.3. Distributions of various quantities for galaxies according to the BGG sSFR class, with BGGs included.

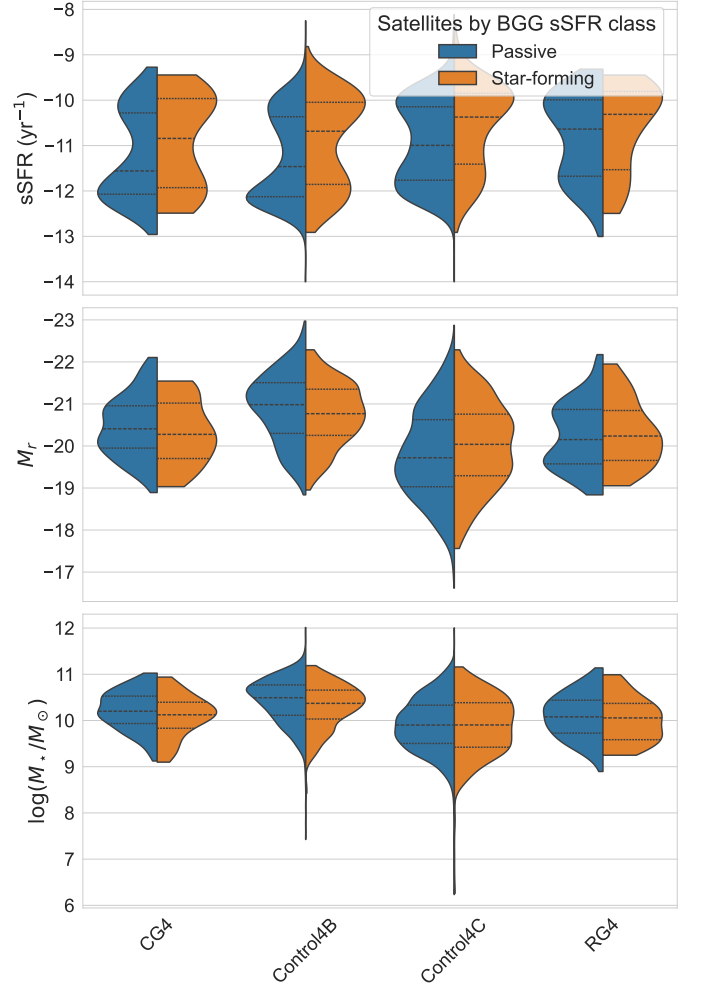


Fig. A.4. Distributions of various quantities for satellites according to the BGG sSFR class.

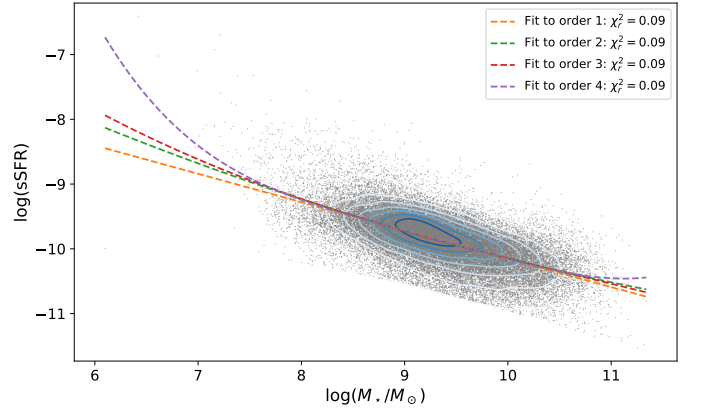


Fig. A.5. Polynomial fits of orders 1–4 to the star-forming main sequence.

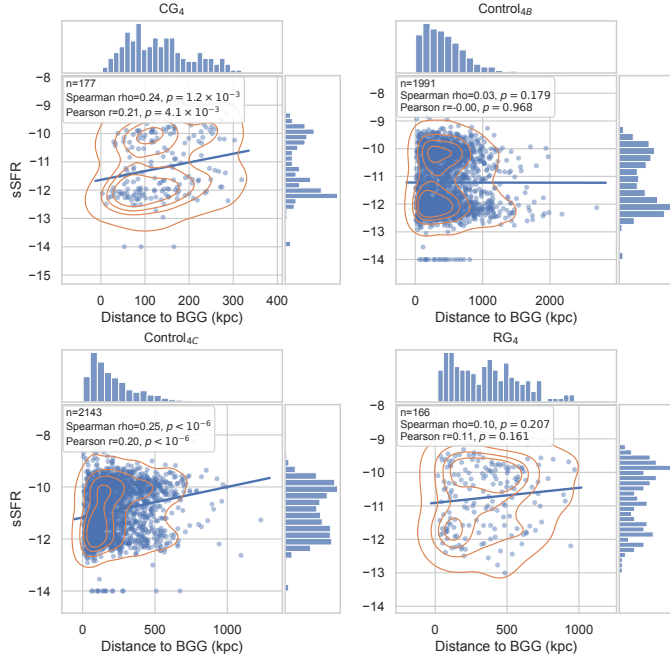


Fig. A.6. sSFR as a function of projected distance to the BGG, in kpc, for satellite galaxies in each sample.

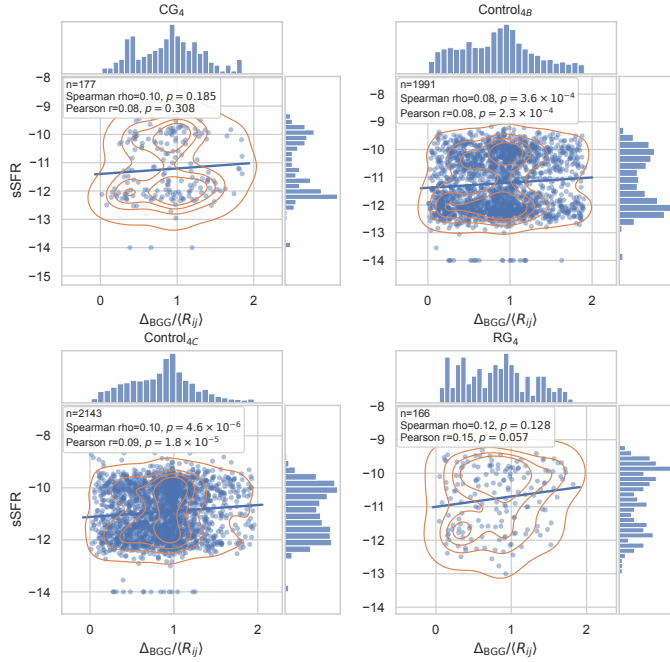


Fig. A.7. sSFR as a function of projected distance to the BGG normalized by $\langle R_{ij} \rangle$ for satellite galaxies in each sample.

A.2. Group-scale and radial diagnostics

960 A.3. Colour diagnostics

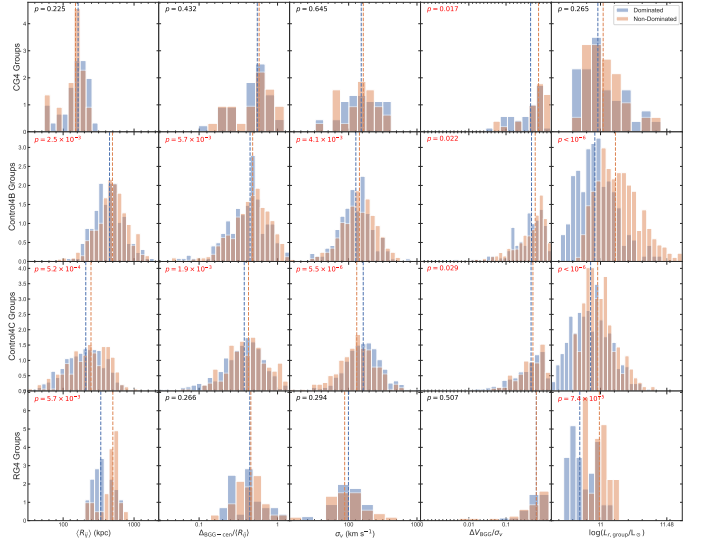


Fig. A.8. Distributions of the main group-scale observables in dominated and non-dominated systems.

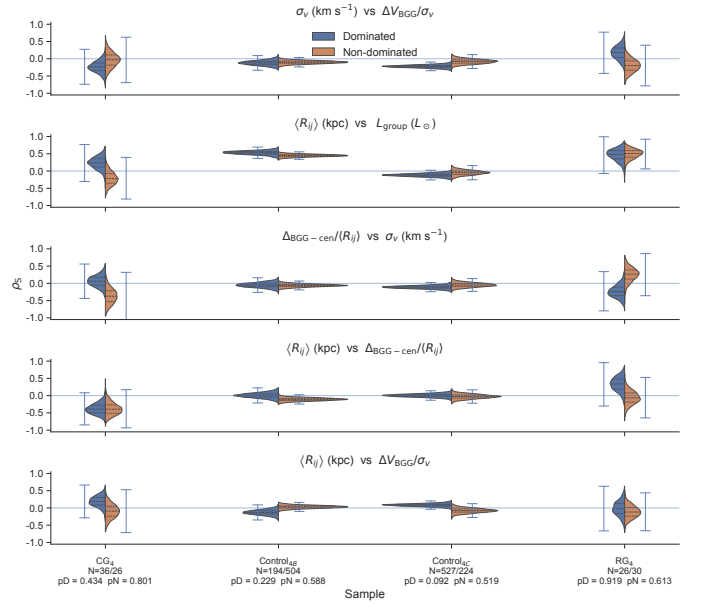


Fig. A.9. Bootstrap distributions of the Spearman coefficients for the group-property pairs selected by the XOR domination criterion.

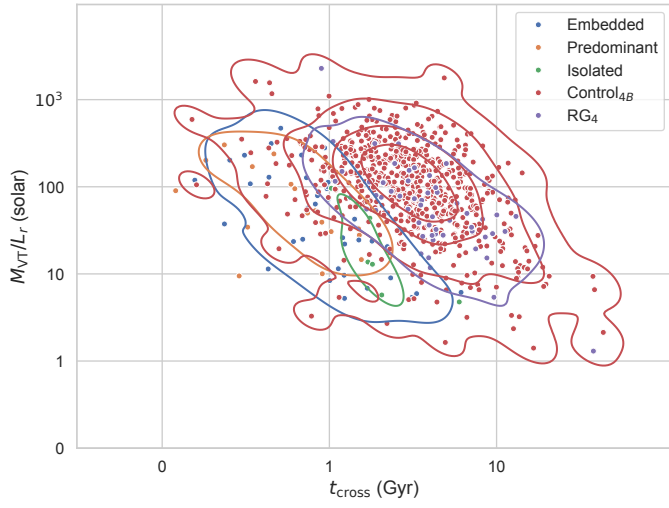


Fig. A.10. Crossing time versus virial mass-to-light ratio for the compact groups and their controls.

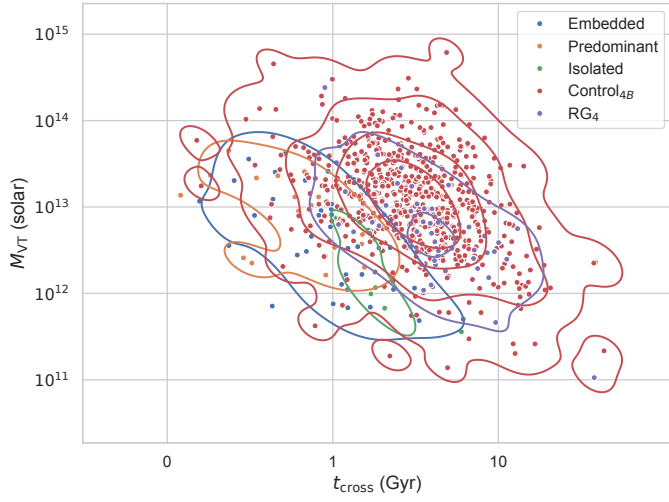


Fig. A.11. Crossing time versus virial mass for the compact groups and their controls.

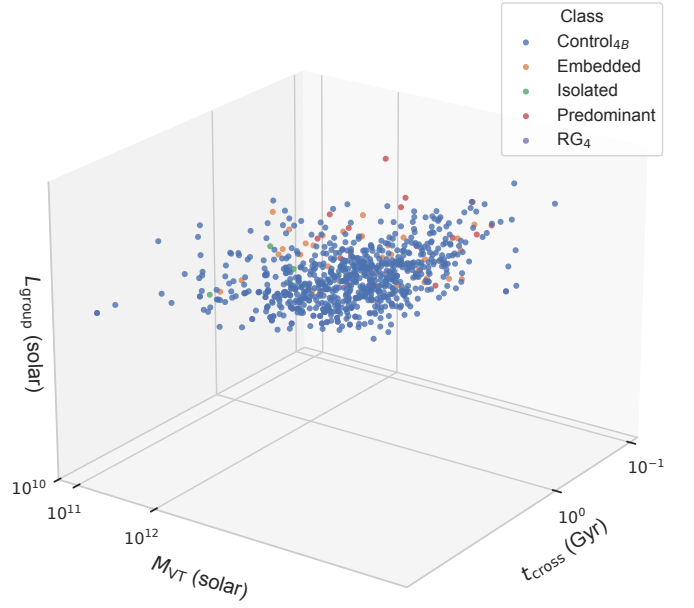


Fig. A.12. Three-dimensional view of the crossing time, virial mass, and group luminosity space.

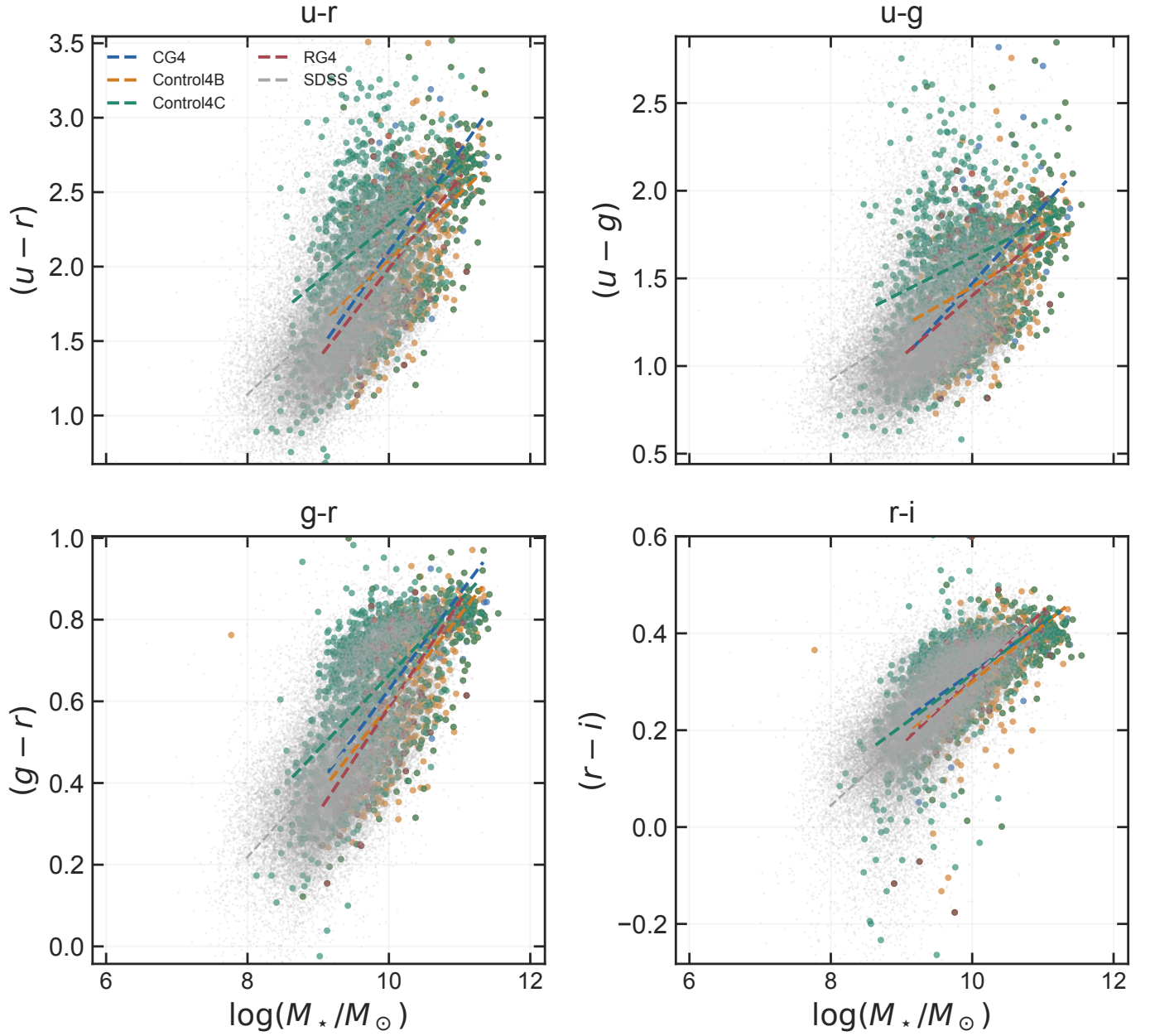


Fig. A.13. Optical colour as a function of stellar mass for CG₄, the three control catalogues, and the matched SDSS reference sample. Dashed lines show the fitted catalogue relations.

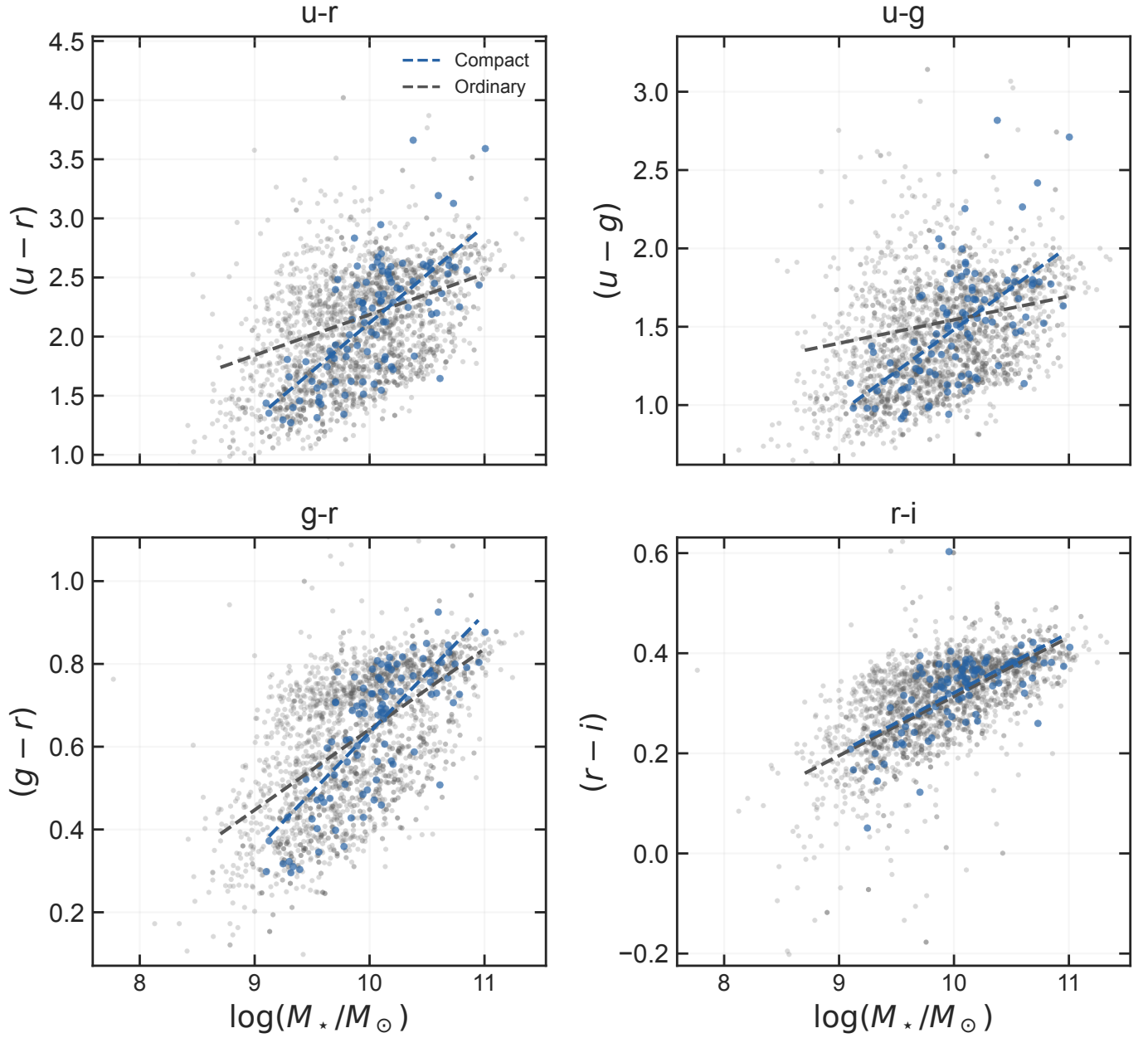


Fig. A.14. Satellite colour–mass relations in compact and ordinary groups. The fitted lines show both colour offsets at the reference mass and differences in slope.

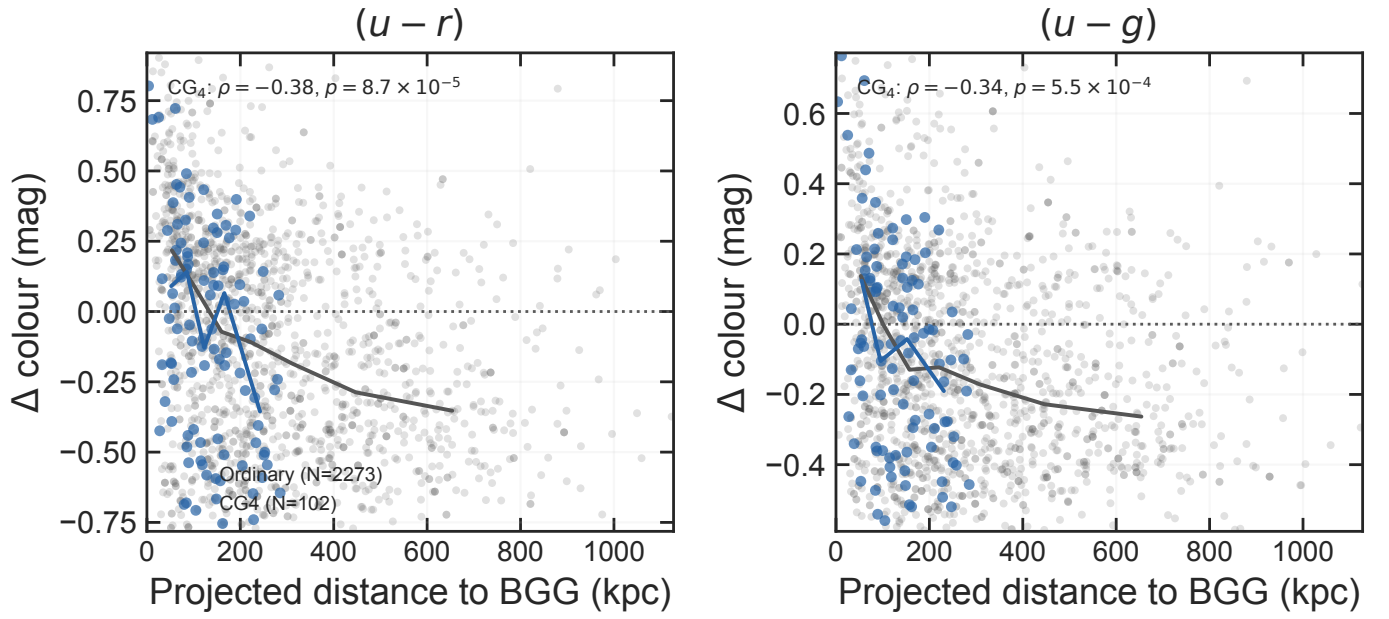


Fig. A.15. Mass- and redshift-adjusted satellite $(u-r)$ and $(u-g)$ residuals as a function of projected distance to the BGG. Lines connect binned medians for CG₄ and the pooled ordinary-group satellites; extreme plotting tails are omitted for readability, while the reported rank correlations use the full matched sample. More negative residuals are bluer than the ordinary-satellite reference relation.

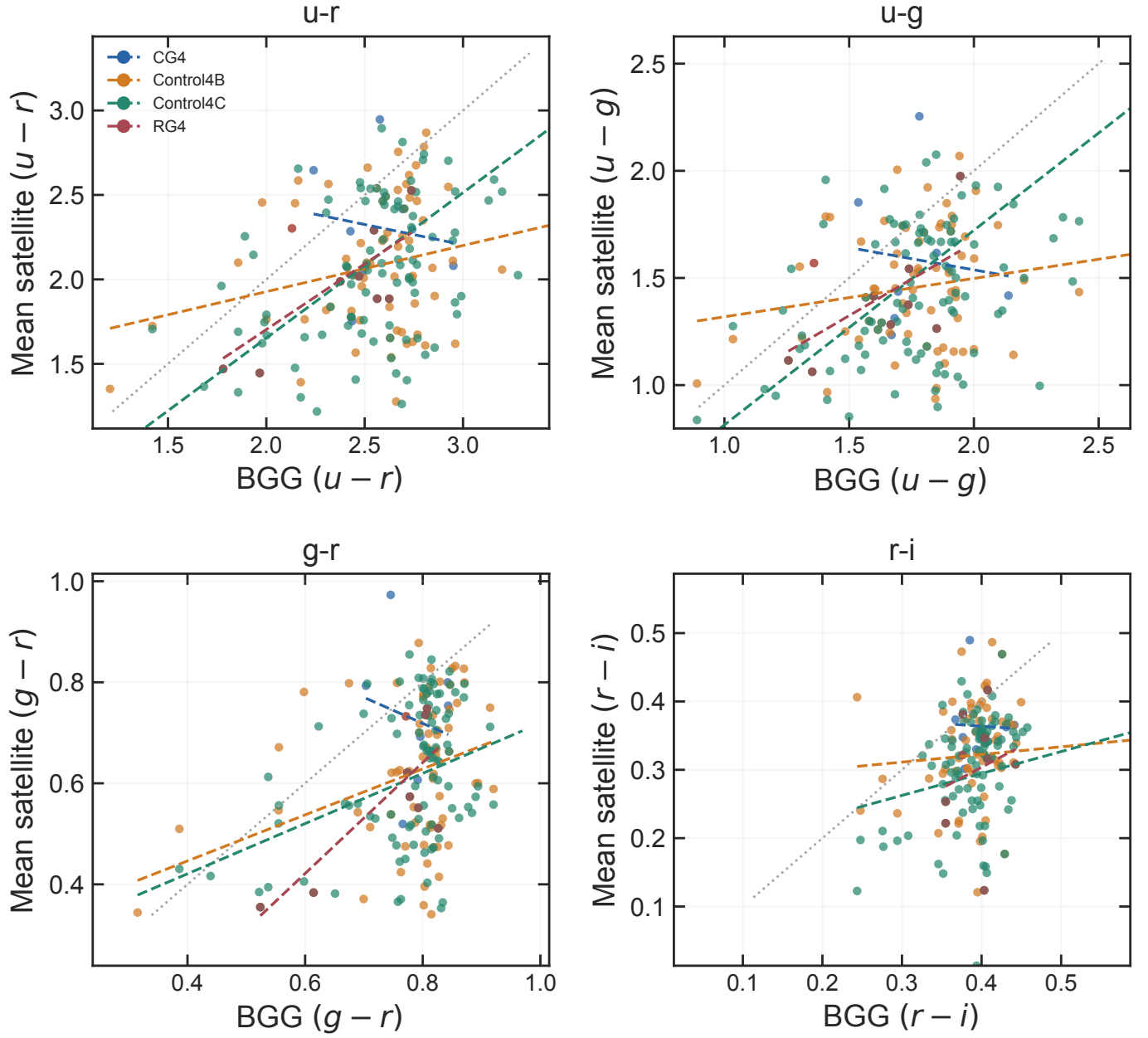


Fig. A.16. Mean satellite colour versus BGG colour for groups with matched photometry. The dotted line is the one-to-one relation and dashed lines are catalogue-specific fits.

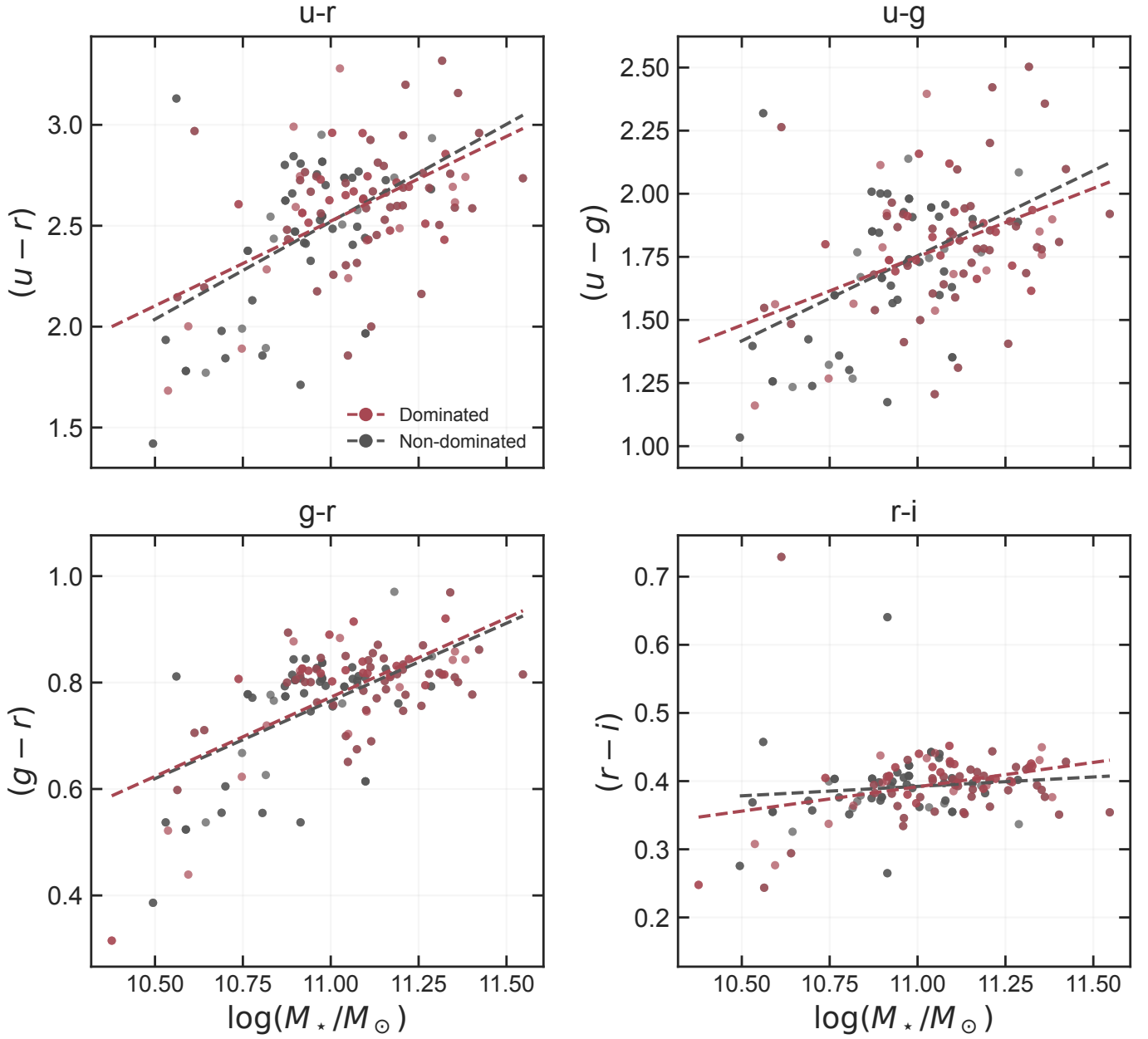


Fig. A.17. BGG colour–mass relations split by luminosity domination. Comparing the fitted relations at a common stellar mass separates colour effects from the different mass distributions.

A.4. Robustness and selection diagnostics

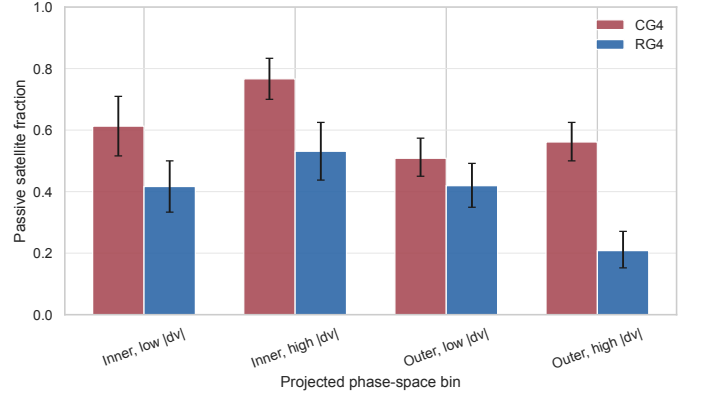


Fig. A.18. Passive satellite fraction in coarse projected phase-space bins. Inner and outer refer to projected-distance rank from the BGG; low and high velocity are split at $|\Delta v_{\text{BGG}}|/\sigma_v = 1$.

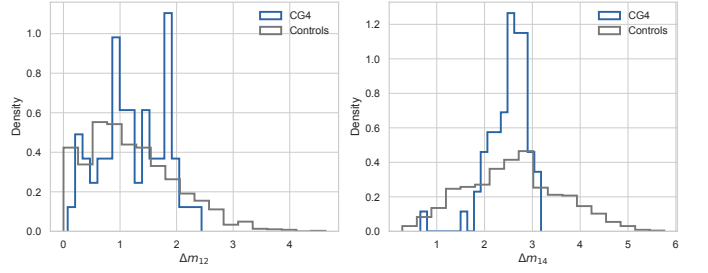


Fig. A.19. Distributions of the first-second and first-fourth magnitude gaps in CG4 and the pooled controls.

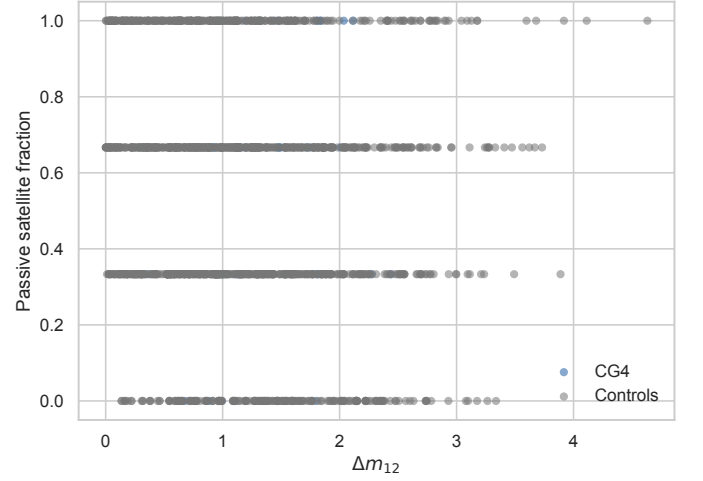


Fig. A.20. Passive satellite fraction as a function of the first-second magnitude gap.

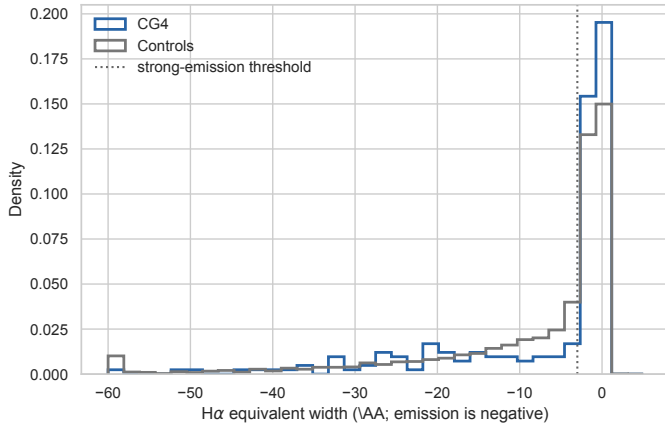


Fig. A.21. H α equivalent-width distributions used for the limited recent-quenching diagnostic.

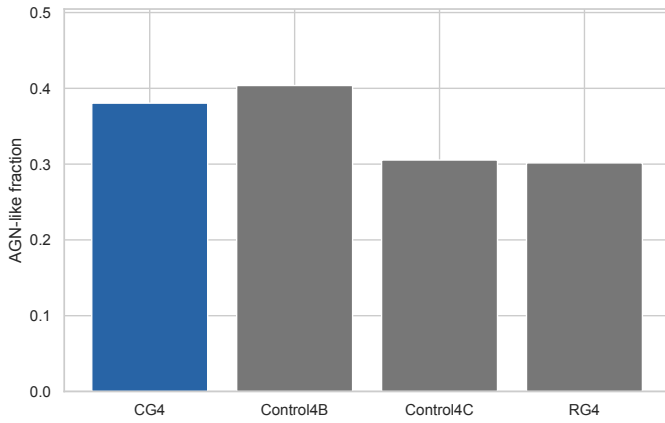


Fig. A.22. AGN-like fractions among galaxies with BPT-classifiable emission-line ratios. BPT-unclassified galaxies are not counted as non-AGN in this diagnostic.

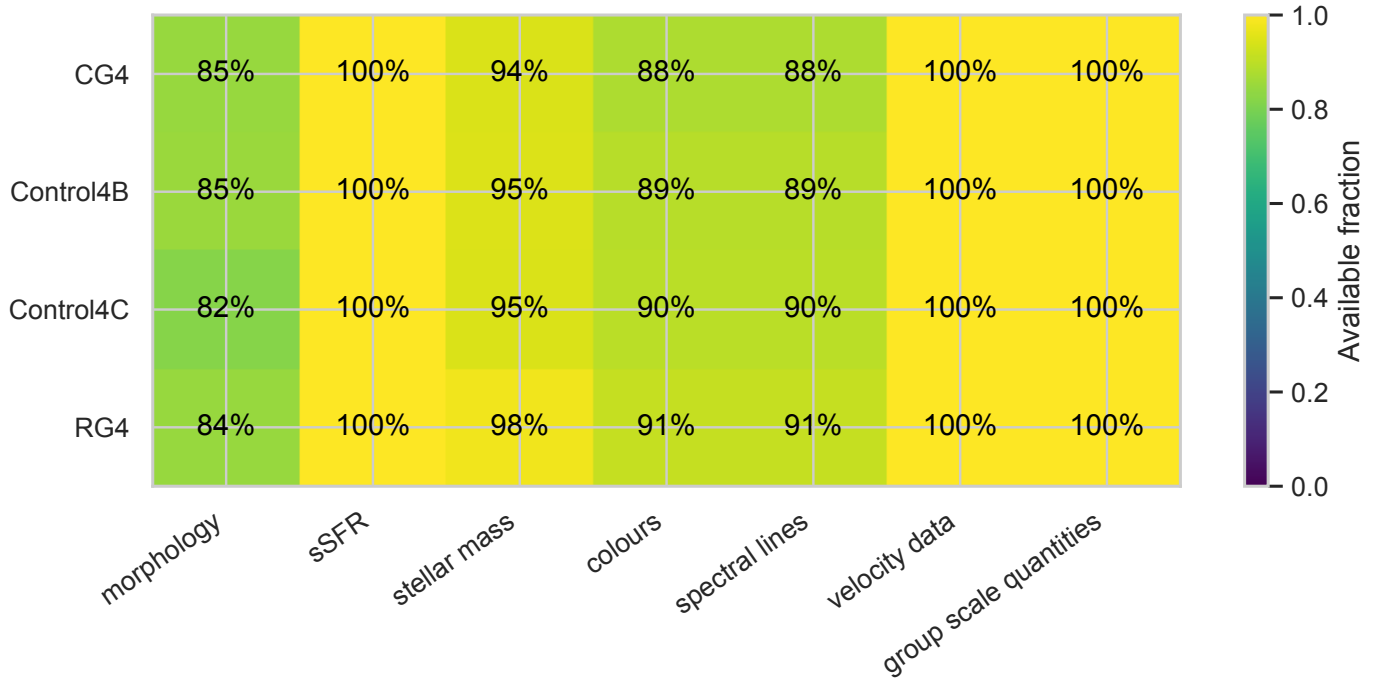


Fig. A.23. Availability fractions for the major derived quantities in each sample.

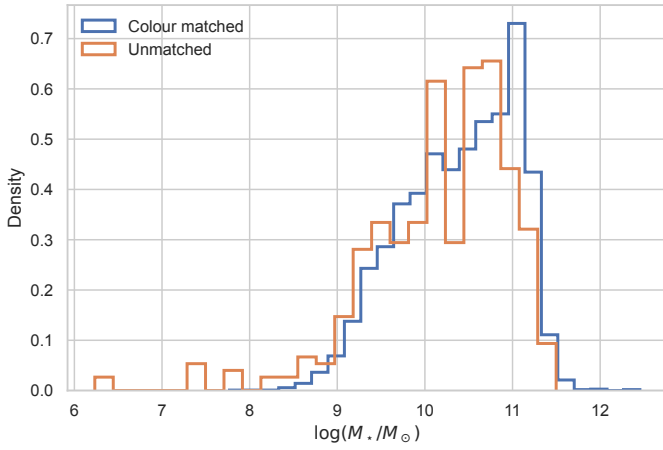


Fig. A.24. Stellar-mass distributions of galaxies with and without complete matched colour measurements.

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